
DISCELL

Anonymous Author
Anonymous Institution

Abstract

A cell’s measured transcriptome in targeted in situ spatial transcriptomics mixes three things: what the cell intrinsically is, what varies with its microenvironment, and transcripts that were assigned to it but belong to its neighbours. The second and third are confounded at first order, since both make a cell’s expression predictable from its neighbours’ types, so no model fit to a single section can tell them apart without further assumptions. We present DISCELL, a variational model that splits a cell’s expression into an intrinsic latent state $\mathbf{z}$, a latent response $\mathbf{w}$ associated with a deterministic description of its microenvironment, and a fixed leakage mixture over graph neighbours governed by a leak fraction κ . Type-conditional invariance keeps niche-predictable information out of $\mathbf{z}$ through an adversarial regulariser, with a closed-form Gaussian penalty as the cheaper first attempt, and is graded by an independently trained held-out probe rather than by the training objective. Because κ is not identifiable from a single section, we do not estimate it: every model is refitted on a grid of κ values, and the deliverable is which spatial effects survive the grid. **[TODO: empirical claims to be added once the evaluation exists]**

1 INTRODUCTION

Imaging-based spatial transcriptomics assays such as Xenium (Janesick et al., 2023) measure thousands of transcripts per cell in situ, at single-cell resolution, over sections containing hundreds of thousands of cells. The appeal is obvious: a cell’s expression can be read together with the identity and state of the cells around it,

which is the raw material for asking how a microenvironment shapes the cells within it. The difficulty is that a cell’s measured count vector is not a clean readout of that cell. It is at least three things superimposed. First, the cell’s *intrinsic* state: its type, its cell-cycle phase, its clone-associated expression, and the systemic signals that reach every cell of its type regardless of position. Second, its *response* to the local microenvironment: the programmes that differ with who its neighbours are and what the surrounding tissue architecture looks like. Third, a *measurement artefact* specific to in situ assays: transcripts that were detected in the vicinity of a segmentation boundary and assigned to the wrong cell. Segmentation of densely packed tissue is imperfect, and published estimates place the fraction of misassigned transcripts anywhere between one tenth and one half **[NEEDS A SOURCE]**.

The second and third components are confounded at first order. A cell whose counts are contaminated by its neighbours acquires their markers; a cell that responds to its neighbours shifts its own programmes in a way their types predict. Where the responding genes are also markers of the neighbouring types, as for many ligands and receptors, or where a programme is shared across a region, the two produce the same signature, and neighbour composition predicts both. That misassigned transcripts produce apparent neighbourhood effects, including spurious cell–cell interaction signals, has been shown directly on in situ data (Mitchel et al., 2026), and the contamination a cell receives grows with the local density of the donor type (Bilous et al., 2026). The two differ in form, one adds a neighbour’s own profile, the other shifts the cell’s own programmes, but from the cell-by-gene count matrix alone the split between them is fixed only by the parametric forms one is willing to assume, and a flexible decoder can mimic either. Any claim that a spatial effect on expression is real therefore depends, silently, on how much of the neighbour-predictable signal was attributed to contamination.

This paper makes that dependence explicit rather than hiding it. We propose DISCELL (*disentangled cell states*), a generative model of per-cell counts with two

latent variables and one fixed mixing operator. A latent $\mathbf{z}_i$ carries the cell’s intrinsic state and is, by architecture and by regularisation, pushed to carry no information about the microenvironment given the cell’s type; a held-out probe measures how much remains. A latent $\mathbf{w}_i$ carries the variation associated with a deterministic context descriptor $\mathbf{c}_i$ that summarises who and what is nearby. A sparse, precomputed leakage kernel β mixes a fraction κ of each neighbour’s clean expression profile into the cell’s expected composition. In the decomposition usual for spatial expression, into intrinsic, spatial, niche, technical and noise terms, the model reads

$$\mathbf{x}_i \mid \ell_i \sim \text{Multinomial}(\ell_i, \mathbf{p}_i),$$

$$\mathbf{p}_i = (1 - \kappa) \text{softmax} \left(\underbrace{a(\mathbf{z}_i)}_{\text{intrinsic}} + \underbrace{\mathbf{B}\mathbf{w}_i}_{\text{niche}} \right) + \underbrace{\kappa \bar{\mathbf{p}}_i}_{\text{leakage}},$$

where $\bar{\mathbf{p}}_i = \sum_j \beta_{ij} \mathbf{p}_j$ is the neighbours’ clean composition. Two technical terms are handled here: the leakage, and the depth ℓ_i , which is conditioned on; the multinomial is the counting noise. The spatial field has no term of its own: variation at the scale of a tissue domain reaches $\mathbf{w}_i$ through the image descriptor in $\mathbf{c}_i$ and is attributed to the niche. Ambient RNA and batch are not modelled, and each fit is one section. The leak fraction κ is not estimated. It is fixed, the model is refitted on a grid of κ values, and every spatial effect is reported as a function of κ with its envelope over random seeds, a sensitivity analysis in which κ is the sensitivity parameter. For each effect we report the smallest leak fraction at which it breaks down: an effect that survives the whole grid cannot be explained by leakage of the modelled form, and one that breaks down at the first leak fraction above zero is not separable even from that. The range across the grid is the deliverable, reported alongside an operating point. It is not a confidence interval, and it does not shrink with more cells.

Our modelling choices follow from three requirements that we motivate in Section 2 and Appendix B. The decoder receives no type label, so that $\mathbf{z}$ is forced to carry identity and cannot degenerate into a residual code. The context encoder queries the neighbourhood with the cell’s type rather than with $\mathbf{z}_i$ and reads only the neighbours’ types, closing a channel through which attention would otherwise reconstruct the cell from its most similar neighbours and make the prior on $\mathbf{w}$ ego-informed. And the invariance we enforce is *conditional* on type: cell types cluster in space, so a marginal invariance between $\mathbf{z}$ and neighbourhood composition deletes cell type from $\mathbf{z}$ altogether, the trade-off known from demographic parity (Zhao and Gordon, 2019) and observed for conditional subspace VAEs (Slavutsky et al., 2026).

Related work. Classical models separate these components by the dependence structure each is assumed to have. Variance-component models decompose each gene’s variance: SpatialDE (Svensson et al., 2018) into a spatially smooth and a non-spatial part with a Gaussian process over coordinates, and SVCA (Arnol et al., 2019) further into intrinsic, environmental and cell–cell interaction components, the last driven by the neighbours’ expression. Nonnegative spatial factorisation (Townes and Engelhardt, 2023) and MEFISTO (Velten et al., 2022) place Gaussian-process priors on latent factors instead. Regression and attribution models tie expression to niche covariates: C-SIDE (Cable et al., 2022) estimates cell-type-specific effects of spatial covariates while modelling the mixture of types within a spot, and MISTy (Tanevski et al., 2022) scores how much of a marker is explained by the cell’s own markers, its immediate neighbours and the wider tissue. Contamination is handled separately, by correcting the counts before any analysis. SoupX, DecontX and CellBender remove ambient RNA (Young and Behjati, 2020; Yang et al., 2020; Fleming et al., 2023). Spot-Clean removes spot swapping on array-based assays with one bleeding rate per slide and a Gaussian distance kernel, both estimated with the help of spots outside the tissue (Ni et al., 2022), which is the closest precedent for our single leak fraction; in situ assays have no such spots, which is one reason we sweep the fraction instead. Baysor segments cells from the transcript positions themselves (Petukhov et al., 2022), and admixture correction removes misassigned transcripts after segmentation (Mitchel et al., 2026). The separation these models achieve rests on assumed dependence structures, which makes it transparent; what they do not provide is a generative model of the cell in which a response to the niche and leakage from it compete for the same counts, and they treat contamination as a preprocessing step rather than as part of the model whose effects are being read.

Node-centric expression models (Fischer et al., 2023) regress a cell’s expression on its neighbourhood directly and produce a niche descriptor without a per-cell latent; our context descriptor $\mathbf{c}_i$ plays that role, and $\mathbf{w}_i$ is what remains after conditioning on it. Several models share our design goal of separating intrinsic from spatially associated variation. SIMVI (Dong et al., 2025) splits a cell’s state into an intrinsic and a spatially induced part with an independence regulariser; MintFlow (Akbarnejad et al., 2025) conditions an intrinsic latent on cell type and a microenvironment latent on neighbour-type composition, and discourages the former from predicting the latter with type-specific discriminators; Celcomen (Megias et al., 2026) separates intra- from inter-cellular gene-interaction programmes with a generative graph neural network; NicheCompass (Birk et al.,

2025) characterises niches with a graph neural network; and Cellina (Moeed et al., 2026) encodes an intrinsic latent from the cell’s counts and an extrinsic one from its neighbours’ expression, ties the former to cell type with a classifier, removes spatial-domain labels from it adversarially, and evaluates the split by predicting expression under a swapped neighbourhood. None of them models transcript leakage, the confound this paper is about, and MintFlow and Cellina name segmentation errors as a limitation [TODO: verify the NicheCompass characterisation; add scVIVA and further spatial VAEs]. Contamination-aware models for in situ data, notably resolVI (Ergen and Yosef, 2025), write a cell’s counts as a mixture of its own decoded expression, its neighbours’ expression decoded by the same network and weighted under a distance-kernel prior, and a slide-wide background, and estimate the mixing and neighbour weights per cell. Our leakage term is the same neighbour-mixing construction with the weights fixed, one leak fraction per section and no background term. We also adopt resolVI’s stop-gradient on the neighbour path, which it uses for training stability, and describe the degeneracy it prevents (Section 2.6). resolVI separates the components by expecting true expression to be low-dimensional. A response to the neighbourhood is low-dimensional too, and the part of it that moves a cell toward its neighbours’ profile has the same effect as leakage, so that expectation no longer separates the two on its own; we therefore sweep the leak fraction rather than estimate it [TODO: support with the planted-leakage synthetic run once the synthetic appendix returns]. Variational disentanglement of condition-specific from shared representations (Klys et al., 2018; Slavutsky et al., 2026) supplies the two-bound objective and the posterior-regularisation view of an adversarial invariance term (Ganchev et al., 2010) that we build on. We depart from them in three ways. The condition is not an observed label but the cell’s niche, summarised by a deterministic context descriptor on which the prior of $\mathbf{w}$ is conditioned. The invariance is conditional on cell type rather than marginal, because type and niche are dependent in tissue, as is standard in fair and domain-invariant representation learning (Madras et al., 2018; Li et al., 2018; Pogodin et al., 2023). And our intrinsic path keeps the full decoder and draws $\mathbf{w}$ from its prior instead of decoding from $\mathbf{z}$ alone (Appendix A.2). Our invariance target is a fixed data descriptor of the niche, so the model cannot shape it to make invariance trivial, unlike an independence penalty between two learned latents (Dong et al., 2025). That adversarial removal is not sufficient without an independent check is the observation of Elazar and Goldberg (2018); our probe protocol is its application. [TODO: extend related work: further spatial VAEs and GNN spatial

models] Table 1 compares the closest models on the properties this paper turns on.

Contributions. (i) A generative model that decomposes in situ counts into intrinsic state, microenvironmental response and neighbour leakage, with a multinomial likelihood in which the leakage mixture is exact under Poisson sources whose leaked-in rate scales with the receiving cell’s own rate rather than the donors’ (Section 2.2). (ii) An inference scheme whose architecture closes information channels that a regulariser cannot reach on its own: the intrinsic encoder never sees the context, the attention context is queried by type, and neighbour quantities are frozen, with an exact two-hop batching scheme that makes the frozen path affordable (Sections 2.3 and 2.6). (iii) A conditional-invariance regulariser, adversarial in its operating configuration with a closed-form penalty as the cheaper first attempt, conditional on type as in the type-specific discriminators of Akbarnejad et al. (2025) but targeting an image-derived niche descriptor alongside neighbour composition, together with a held-out probe protocol that grades either independently of the training objective (Section 2.5). (iv) The leakage sweep as a sensitivity analysis: spatial effects are reported as trajectories over κ with seed envelopes, alongside an operating point (Section 2.8). [TODO: (v) the empirical evaluation, to be added]

2 THE DISCELL MODEL

We present the model in the order in which its parts depend on one another: the observed quantities and the spatial graph (Section 2.1); the generative model and the assumptions behind it (Section 2.2); the inference networks (Section 2.3); the objective (Section 2.4); the conditional-invariance regulariser and the held-out probe that grades it (Section 2.5); the frozen-neighbour batching that makes the model trainable (Section 2.6); model selection (Section 2.7); and what is and is not identified, which is what turns a fit into a finding through the leakage sweep (Section 2.8). Each design decision is stated with its reason; the longer arguments are in Appendix B and the derivations in Appendix A. Figure 1 summarises the model for one cell.

2.1 Setting and Notation

We consider a single tissue section profiled with a targeted in situ assay. For each of N segmented cells i we observe a count vector $\mathbf{x}_i \in \mathbb{N}^G$ over G genes, its total $\ell_i = \sum_g x_{ig}$, a cell-type label $t_i \in \{1, \dots, K\}$, and the morphology image around the cell. The label may be a curated annotation or a frozen unsupervised clustering and is treated as data, with two caveats that recur be-

Table 1: The models closest to DISCELL, compared on the properties this paper turns on. *Intrinsic latent*: a per-cell latent for the cell’s own state. *Niche term*: a component of the model that depends on the neighbourhood. *Leakage*: misassigned transcripts are part of the generative model; *leak share*: how the amount of leakage is set. *Invariance*: the regulariser that keeps the intrinsic latent free of the niche, and whether it conditions on type. *Counterfactuals*: expression predicted under an intervention. ✓: yes; ×: no; –: not applicable.

Model	Intrinsic latent	Niche term	Leakage	Leak share	Invariance	Counterfactuals
resolVI (Ergen and Yosef, 2025)	✓	×	✓	estimated per cell	×	×
SIMVI (Dong et al., 2025)	✓	spatially induced latent	×	–	independence, marginal	×
MintFlow (Akbarnejad et al., 2025)	✓	microenvironment latent	×	–	adversarial, given type	in silico perturbation
Celcomen (Megas et al., 2026)	×	inter-cellular gene interactions	×	–	–	gene knockouts
Cellina (Moeed et al., 2026)	✓	extrinsic latent	×	–	adversarial against domain labels, marginal	swapped neighbourhoods
DISCELL	✓	context-conditional response $\mathbf{w}_i$	✓	one per section, swept	adversarial, given type	a changed context, with abduction

low: it was itself derived from the same contaminated counts, and its granularity is a modelling choice that sets what the decomposition is relative to (Section 2.5). Cells without a label form one additional class. In training it enters every type-conditioned quantity like any other, and its cells remain neighbours of the cells around them. It is never a target of evaluation: every diagnostic and readout of Sections 2.7 and 2.8 is computed on labelled cells only, because a residual class is not a cell type, and a read that treated it as one would measure the failure of the label rather than the model. Curated annotations frequently encode state or location as well as lineage (tumour- versus stroma-associated fibroblasts, hypoxic or proliferating tumour subsets). Such labels are niche-informed: conditioning on them assigns the corresponding response to t , where neither $\mathbf{w}$ nor the invariance of Section 2.5 sees it. Labels should therefore be lineage-level, with state-bearing subtypes merged before fitting, as they are in every fit reported here. The total ℓ_i is conditioned on throughout and never modelled.

Spatial graph. Cells are nodes; two cells are adjacent if and only if their Voronoi cells share a face, i.e. no third cell lies between them (Delaunay adjacency). This choice has no neighbourhood count k and no radius, adapts to local packing with an average degree near six, and uses only cell centroids, so it is insensitive to segmentation-boundary noise and to the nuclear-expansion segmentation that in situ platforms fall back on. Delaunay adjacency also connects cells across lumens, vessels and tears, so edges longer than

a cut-off ($40\ \mu\text{m}$ here) are pruned. We write $\mathcal{N}_i$ for the pruned neighbour set of i , which excludes i itself; all its members are first-order neighbours, so no cell-size correction of distances is needed. The same graph serves every role below.

For each edge we record the centroid distance d_{ij} and the length f_{ij} of the shared Voronoi face, measured after clipping each Voronoi cell to a $30\ \mu\text{m}$ disc around its centroid so that faces at tissue borders and across gaps are bounded. The face length is a proxy for contact extent that is defined whether or not the segmented polygons touch. We deliberately do not use the length of shared segmented boundary: segmented polygons of adjacent cells rarely touch, so a contact length would vanish on a large share of edges, and where it is non-zero it tracks the gap between polygons, which is local density, a property of the niche that we want to keep out of the *technical* part of the model, the part that describes measurement rather than biology (Appendix B.2; Figure 3 and Table 3).

Derived observables. From the graph and labels we form the neighbour-type composition $\mathbf{y}_i \in \Delta^{K-1}$, $y_{ik} = |\{j \in \mathcal{N}_i : t_j = k\}|/|\mathcal{N}_i|$, which excludes the cell itself. From the image we form an embedding $\Phi_i \in \mathbb{R}^{d_\Phi}$ of a patch centred on the cell, produced by a frozen pretrained vision model, in which a disc covering the cell has been masked out, so that Φ_i describes the surrounding tissue and not the cell (Appendix B.3). The two descriptors act at different scales: the graph is one hop, a few to thirty micrometres, while the

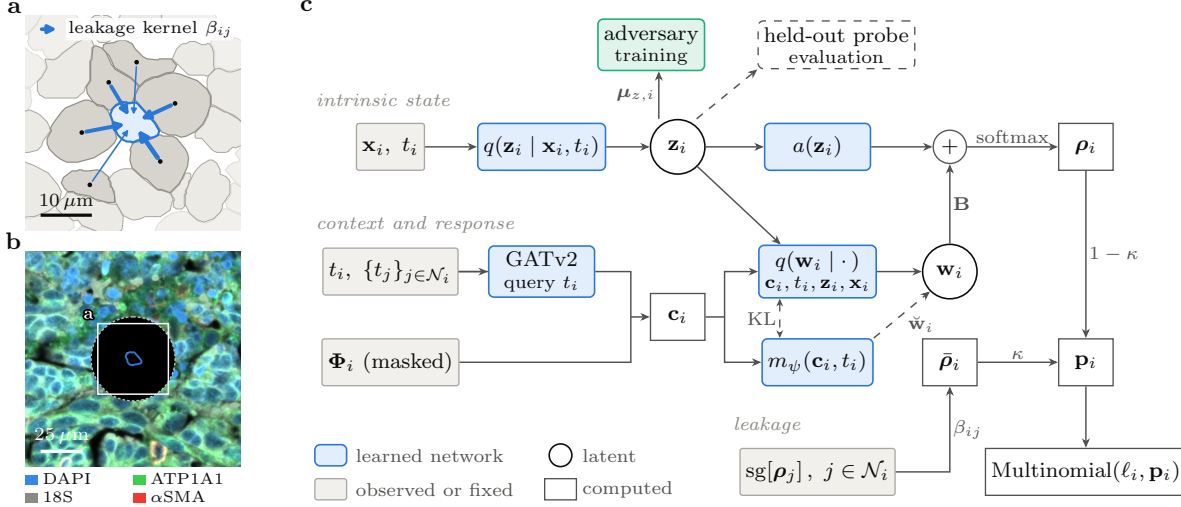


Figure 1: DISCELL for one cell i . **a**, The leakage kernel on the primary section: the cell (blue), its first ring of neighbours in the pruned Delaunay graph (dark grey), each sending an arrow whose width grows with its weight β_{ij} , and its second ring (light grey), which the batching of Section 2.6 also reads. The attention that builds the context runs over the same first ring but uses no edge features: its weights depend on the two cells' types alone (Equation (27)). **b**, The input of the image descriptor Φ_i : the four-channel morphology crop the image model receives (128 μm field; the markers DAPI, ATP1A1, 18S and αSMA as the image model reads them, shown as one colour composite; on the slide the ATP1A1 channel is a membrane mix with CD45 and E-cadherin, and the αSMA channel a mix with vimentin), with the 25 μm disc over the cell set to zero; the cell's outline is drawn for reference. The square marked a is the window of panel a: the zeroed disc covers the cell and most of its first ring, so the descriptor describes the tissue around the leakage neighbourhood rather than the neighbourhood itself. Tissue panels are in image orientation. **c**, The computation graph. The intrinsic state $\mathbf{z}_i$ is encoded from the cell's own counts and type. The context $\mathbf{c}_i$ combines attention over the neighbours' types, queried with the cell's type, with Φ_i . The response $\mathbf{w}_i$ has the prior $p(\mathbf{w}_i | \mathbf{c}_i, t_i)$ with mean m_ψ , and a posterior that also sees the cell. The clean composition $\rho_i = \text{softmax}(a(\mathbf{z}_i) + \mathbf{B}\mathbf{w}_i)$ is mixed with the neighbours' clean compositions, held under a stop-gradient, through β and the fixed leak fraction κ . The dashed $\tilde{\mathbf{w}}_i$ is the prior draw used by the intrinsic path. The adversary acts on the posterior mean $\mu_{z,i}$ during training; the held-out probe grades it afterwards and never enters training.

masked patch describes tissue beyond roughly 25 μm ; both are what we mean by “the niche”. Here d_Φ is the embedding's native width; whether a low-dimensional projection should replace it is an open implementation choice (Appendix C.2). Three lookup tables are fixed before training: $\bar{\mathbf{y}}(t)$, the mean composition of cells of type t ; $\bar{\Phi}(t)$, the mean image-niche membership of type t ; and $\bar{\mathbf{v}}(t)$, the mean of the invariance block of Section 2.5 for type t . Isolated cells, those with no neighbours after pruning, have an empty composition and are excluded from all three tables.

Leakage kernel. The leakage kernel is a sparse matrix over the same edges, normalised over the neighbours of each connected cell,

$$\beta_{ij} \propto f_{ij} \exp(-d_{ij}/\tau), \quad \sum_{j \in \mathcal{N}_i} \beta_{ij} = 1, \quad (1)$$

with contamination decay length $\tau = 20 \mu\text{m}$; it is row-stochastic on connected cells, and an isolated cell keeps

an all-zero row. It is precomputed once and never learned, which is where it departs from the neighbour-mixing term of Ergen and Yosef (2025), whose weights are fitted per cell. Continuous geometry is allowed here, and only here: β is not part of any conditioning variable and does not touch the estimand (Appendix B.2).

2.2 Generative Model

Each cell has two latent variables. The intrinsic state $\mathbf{z}_i \in \mathbb{R}^{d_z}$ carries what the cell is; the response $\mathbf{w}_i \in \mathbb{R}^{d_w}$ carries how cells of that type shift with their surroundings. The environment itself enters as a deterministic context descriptor $\mathbf{c}_i$ (defined in Section 2.3) that is shared by co-located cells. With m_ψ a learned prior-mean network, a a learned decoder, $\mathbf{B} \in \mathbb{R}^{G \times d_w}$ a learned loading matrix and $\kappa \in [0, 1]$ the fixed leak

fraction,

$$\mathbf{z}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (2)$$

$$\mathbf{w}_i \mid \mathbf{c}_i, t_i \sim \mathcal{N}(m_\psi(\mathbf{c}_i, t_i), \sigma_w^2 \mathbf{I}), \quad (3)$$

$$\boldsymbol{\rho}_i = \text{softmax}(a(\mathbf{z}_i) + \mathbf{B}\mathbf{w}_i), \quad (4)$$

$$\bar{\boldsymbol{\rho}}_i = \sum_{j \in \mathcal{N}_i} \beta_{ij} \boldsymbol{\rho}_j, \quad (5)$$

$$\mathbf{p}_i = (1 - \kappa) \boldsymbol{\rho}_i + \kappa \bar{\boldsymbol{\rho}}_i, \quad (6)$$

$$\mathbf{x}_i \mid \ell_i \sim \text{Multinomial}(\ell_i, \mathbf{p}_i). \quad (7)$$

Here $\boldsymbol{\rho}_i \in \Delta^{G-1}$ is the cell's own *clean* composition, $\bar{\boldsymbol{\rho}}_i$ is the *foreign influx*, the leakage-weighted average of its neighbours' clean compositions, and $\mathbf{p}_i$ is the fitted multinomial probability. The decoder $a(\mathbf{z}_i) \in \mathbb{R}^G$ is the cell's baseline programme in log space; column k of $\mathbf{B}$ is a gene programme and w_{ik} is how far cell i has moved along it. The prior variance is fixed at $\sigma_w = 1$, deliberately (Section 2.3).

For an isolated cell $\bar{\boldsymbol{\rho}}_i = \mathbf{0}$, and the mixture in Equation (6) would sum to $1 - \kappa$, charging the cell $-\ell_i \log(1 - \kappa)$ nats for leakage it cannot receive. The mixture is therefore renormalised per row,

$$\mathbf{p}_i = \frac{(1 - \kappa) \boldsymbol{\rho}_i + \kappa \bar{\boldsymbol{\rho}}_i}{(1 - \kappa) + \kappa \sum_{j \in \mathcal{N}_i} \beta_{ij}}, \quad (8)$$

so that an isolated cell degenerates to $\mathbf{p}_i = \boldsymbol{\rho}_i$, i.e. $\kappa_i = 0$: no neighbours, no foreign source. For connected cells the denominator is one. Probabilities are floored at 10^{-8} inside the logarithm. Isolated cells keep an all-zero composition $\mathbf{y}_i$ and are not excluded from the regulariser or the probes of Section 2.5.

What the leakage term assumes. The leak fraction κ is section-wide, not per cell, and there is no ambient term. A cell's contamination in reality varies with its own size and depth, the depth and density of its neighbours, and how much boundary they share. The model scales the leaked rate with the receiving cell's own rate (below), so a low-count cell beside high-count donors is under-corrected at every κ , and a high-count cell beside low-count donors over-corrected. The invariance of Section 2.5 keeps $\mathbf{z}$ free of anything niche-predictable, so the niche-predictable part of that error is absorbed by $\mathbf{w}$ and is not addressed by the sweep. Labels are derived from uncorrected counts and held fixed across the grid. Leakage-driven label errors concentrate where leakage is largest, at heterotypic interfaces and in low-count cells beside high-count neighbours; for such a cell the error enters through t_i and through its neighbours' compositions, where neither κ nor the invariance reaches it. Diffuse background is assumed negligible relative to leakage from adjacent cells [NEEDS A SOURCE]. The mixture in Equation (6) is exact under two assumptions: that the sources are independent

Poisson variables, whose sum is again Poisson, and that the rate of leaked-in transcripts scales with the receiving cell's own rate rather than with the donors', which is where the per-cell variation just mentioned is hidden. The multinomial then follows from conditioning on the observed total (below). The parametrisation also assumes that leakage is gene-agnostic: κ carries no gene index, so a leaked transcript is drawn from the donor's whole-cell composition $\boldsymbol{\rho}_j$. Transcripts nearest the boundary, such as mRNAs of secreted and membrane proteins, are plausibly over-represented in what crosses it [NEEDS A SOURCE]. Such a gene-specific leak is not spanned by κ at any value and lands in $\mathbf{w}$, so a response programme enriched for cell-surface genes cannot be separated from leakage by the sweep alone.

What the decomposition means. The split between $\mathbf{z}$ and $\mathbf{w}$ is defined by two requirements. First, within a type, the intrinsic state should carry nothing that is predictable from the niche: $\mathbf{I}(\mathbf{z}_i; [\mathbf{y}_i, \boldsymbol{\Phi}_i] \mid t_i) = 0$. In Equation (2) $\mathbf{z}_i$ is a priori independent of everything, and $\mathbf{y}_i, \boldsymbol{\Phi}_i$ are not modelled, so this is not a property of the generative model; it is a constraint imposed on the variational family in Section 2.5, in the sense of posterior regularisation (Ganchev et al., 2010). Second, the environment reaches the counts only through the response and through leakage: $\mathbf{x}_i \perp \mathbf{c}_i \mid \mathbf{z}_i, \mathbf{w}_i, \{\boldsymbol{\rho}_j\}_{j \in \mathcal{N}_i}$, which holds by construction because the decoder of Equation (4) takes no context. Precisely, what the first requirement removes from $\mathbf{z}$ is variation predictable from the neighbour composition and the masked image given the type. It removes intrinsic variation that is spatially organised within a type only to the extent that neighbour composition and the masked image predict it. On a carcinoma section that extent can be large: subclones occupy contiguous regions and the masked image shows the nuclei of clonal kin, so part of a cell's clone-associated expression is predictable from its niche, is moved from $\mathbf{z}$ into $\mathbf{w}$, and would be read as a response; the part they do not predict stays in $\mathbf{z}$, as does variation carried by the neighbours' own intrinsic codes. How much is at stake depends on the labels as well: a label that splits subclones keeps them in t , while under the lineage-level labels recommended above they are within-type variation, and on a carcinoma section this is then the main case rather than a corner case. This is why the decomposition is relative to t (Section 2.5).

Why a multinomial. If $x_{ig} \sim \text{Poisson}(\lambda_{ig})$ independently, then $\mathbf{x}_i \mid \sum_g x_{ig} = \ell_i$ is $\text{Multinomial}(\ell_i, \boldsymbol{\lambda}_i / \sum_g \lambda_{ig})$. Since ℓ_i is observed, the multinomial is the coherent likelihood: no library-size latent and no double counting of depth. Poisson sources are closed under addition, which is what makes the

leakage mixture exact; negative-binomial sources with gene-specific means are not in general, and a negative binomial on the mixed counts would let the dispersion absorb part of what the mixture should explain. At the depth of in situ data, tens to a few hundred counts over thousands of genes, gene-wise overdispersion cannot be estimated in any case, and a low-dimensional amortised $\mathbf{z}$ absorbs much of what a dispersion parameter captures in coarser models. Should posterior predictive checks show overdispersion at higher depth, the appropriate extension is a Dirichlet-multinomial with one concentration scalar, not G dispersions.

Why κ is fixed. Leakage and response differ in form: one adds a neighbour’s own profile through a fixed contact kernel, the other shifts the cell’s own programmes along learned directions. From the cell-by-gene count matrix alone, however, the split between them is pinned only by those parametric forms, and a decoder flexible enough to fit the data can mimic either, since neighbour composition predicts both. The likelihood alone cannot choose κ : the data bound it from above but not from below (Proposition 2). Instead we fix it, fit the model independently at every point of a grid, and report each readout as a function of κ (Section 2.8). $\kappa = 0$ is the nested no-correction null. The grid is anchored at zero and extends to 0.4, close to the upper range of published misassignment estimates for this platform [NEEDS A SOURCE]; a data-driven upper bound from an atlas-defined marker set, and the transcript-level information that would identify κ from a nuclear/extranuclear split of the counts, are discussed in Appendix B.6.

Why the decoder has no type input. The decoder in Equation (4) takes $\mathbf{z}_i$ and $\mathbf{w}_i$ only, so the only route from identity to expression runs through $\mathbf{z}$. If $\mathbf{z}$ collapsed to a function of the label, both reconstruction terms in Section 2.4 would lose all within-type variance, a large fitting penalty, and that is what holds $\mathbf{z}$ open. With the label in the decoder, $\mathbf{z}$ would become a residual-only code and would no longer be interpretable as identity. What $\mathbf{z}$ carries beyond the label is what the label discretises away: cell-cycle phase, clone-associated expression, position on a differentiation continuum, stress and metabolic state, states acquired in a niche the cell has since left, and systemic signals that affect every cell of a type regardless of position. The last is environmental in the colloquial sense but not niche-predictable, and it belongs in $\mathbf{z}$. Two remarks follow. Since the encoder of Section 2.3 receives the label, the agreement between $\mathbf{z}$ and t measures whether $\mathbf{z}$ retains type, not whether it discovers it; its collapse is the detector for the marginal-invariance failure mode (Section 2.7). And coarse labels leave more for $\mathbf{z}$ to

Table 2: Learned and fixed quantities. $d_z = 20$, $d_w = 6$; G , K and d_Φ are set by the panel, the label set and the image model.

Learned	Fixed before training
encoder of $q(\mathbf{z}_i \mathbf{x}_i, t_i)$; attention layer and type query embed(t); prior mean m_ψ ; encoder of $q(\mathbf{w}_i \cdot)$; decoder a ; loadings $\mathbf{B}$; adversary heads (separate optimiser)	graph, kernel β , τ ; leak fraction κ (swept); prior scale $\sigma_w = 1$; image embedding Φ and its principal components; image-niche memberships e^Φ ; tables $\bar{\mathbf{y}}(t)$, $\Phi(t)$, $\bar{\mathbf{v}}(t)$; labels t ; totals ℓ

carry; whether $\mathbf{z}$ separates cell-cycle phases within a type is an empirical check for the evaluation.

2.3 Inference Networks

Table 2 summarises what is learned and what is fixed. All networks are multilayer perceptrons unless stated; widths are in Table 5.

The context descriptor $\mathbf{c}_i$. One vector per cell summarising who and what is nearby. It is deterministic, is shared by co-located cells, and is the conditioning variable of the model. It is computed by a single graph-attention layer (Brody et al., 2022) over the pruned graph, concatenated with the image embedding and an isolation flag:

$$\mathbf{h}_j = \text{onehot}(t_j), \quad (9)$$

$$\mathbf{g}_i = \text{GATv2}(\text{query} = \text{embed}(t_i), \{\mathbf{h}_j\}_{j \in \mathcal{N}_i}), \quad (10)$$

$$\mathbf{c}_i = \mathbf{g}_i \oplus \Phi_i \oplus \mathbf{1}[\mathcal{N}_i = \emptyset]. \quad (11)$$

Here $\text{embed}(t_i)$ is a learned per-type vector of width K , the width of the source features; since the layer applies its own linear map to the query, this is equivalent to a one-hot query and is a parameterisation, not a design choice. The layer has no self-loops, no edge features, and its heads are averaged. Its attention weights π_{ij} satisfy $\sum_j \pi_{ij} = 1$. The context is therefore a pure function of data, types, graph and image; no latent enters it. With types as both query and sources, the logit of an edge depends on the two types alone, so the attention output is a learned, type-specific reweighting of the neighbour composition, and $\mathbf{c}_i$ is a fixed function of $(t_i, \mathbf{y}_i, \Phi_i)$: the descriptors against which $\mathbf{z}$ is made invariant, so the prior on $\mathbf{w}$ sees exactly what $\mathbf{z}$ is kept from (Appendix B.2). Three decisions here are load-bearing.

The query is the type, not $\mathbf{z}_i$. In attention of this form the destination feature enters the attention logits but never the aggregated value, so using $\mathbf{z}_i$ as the query would inject no ego content directly. It would, however, open a selection channel: attention can learn

$\pi_{ij} \propto \exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_j))$ and reconstruct the cell through its most similar neighbours, so that $\mathbf{c}_i$ becomes a mirror of $\mathbf{z}_i$. The incentive exists because an ego-informed prior on $\mathbf{w}_i$ is free decoder capacity, and it is strongest in homotypic regions, where κ is also least identified. The consequence would be that m_ψ already knows the cell, the divergence $\text{KL}(q(\mathbf{w}_i) \| p(\mathbf{w}_i | \mathbf{c}_i, t_i))$ collapses, and $\mathbf{w}_i$ measures deviation from *self* rather than from the expected response. Querying with the type closes the selection channel at no cost to the prior, since $m_\psi(\mathbf{c}_i, t_i)$ already receives t_i . The sources are the neighbours' types only. An earlier version also fed the neighbours' posterior means under a stop-gradient; removing them improved held-out reconstruction, type agreement and the mirror diagnostic in a three-seed ablation while leaving the response's spatial readouts unchanged, so the input bought nothing measurable and was dropped. The within-type mirror coefficient is still monitored rather than assumed away (Section 2.7; Appendix B.1).

No per-edge geometry enters the context. The graph says who is adjacent; the image describes how packed and how structured the region is; per-edge distances and contact extents are withheld from $\mathbf{c}_i$. Three reasons, in decreasing order of weight. The concrete risk is collinearity: an attention layer given edge features can rediscover the leakage kernel β , after which $\mathbf{w}$ becomes collinear with $\tilde{\rho}$ and the two mechanisms the sweep is meant to separate merge inside the model (Appendix B.2). Within a first-order neighbourhood the distances span perhaps 5 to 30 μm , most of which is local density, which the image already carries. And the intended reading of the context is that neighbour-type composition is the exposure of interest while the image and the graph are conditioning covariates; a hypothetical change in composition is a meaningful question about a tissue in a way that “the same neighbours, further away” is not, and putting per-edge geometry into the conditioning variable would mix the two. The estimand is used only informally here; counterfactual machinery is future work (Section 4).

Isolated cells. A cell with no in-edges has an empty attention set; the graph part of $\mathbf{c}_i$ is set to zero and the flag is raised. The image part stays, since Φ_i is measured data defined for every cell. Together with Equation (8), an isolated cell is modelled with $\kappa_i = 0$ and an image-only context.

Posterior over $\mathbf{z}$. The posterior $q(\mathbf{z}_i | \mathbf{x}_i, t_i)$ is a diagonal Gaussian produced by a network whose input is the library-normalised, log-transformed count vector together with the log total and the one-hot type,

$$[\log(1 + \mathbf{x}_i \tilde{\ell} / \ell_i), \log \ell_i, \text{onehot}(t_i)], \quad \tilde{\ell} = \text{median}_i \ell_i. \quad (12)$$

The encoder *never sees $\mathbf{c}_i$* : invariance by architecture, which the regulariser of Section 2.5 then completes. Library normalisation before the log matters because a raw $\log(1 + \mathbf{x})$ entangles depth with composition, so that a 300-count and a 50-count cell of the same type look very different; since ℓ_i is conditioned on rather than modelled, $\mathbf{z}$ should not spend capacity on it. Total counts are nonetheless weakly informative about state, so $\log \ell_i$ enters as one extra scalar rather than being discarded.

Prior and posterior over $\mathbf{w}$. The prior is what a typical cell of this type would do in this context; the posterior is what this cell did:

$$p(\mathbf{w}_i | \mathbf{c}_i, t_i) = \mathcal{N}(m_\psi(\mathbf{c}_i, t_i), \sigma_w^2 \mathbf{I}), \quad (13)$$

$$q(\mathbf{w}_i | \mathbf{c}_i, t_i, \mathbf{z}_i, \mathbf{x}_i) = \mathcal{N}(\boldsymbol{\mu}_{w,i}, \text{diag } \boldsymbol{\nu}_{w,i}^2), \quad (14)$$

with $\boldsymbol{\nu}_{w,i}$ the learned posterior scale, as distinct from the fixed prior scale σ_w . Both distributions are functions of the context, but only the posterior sees the cell itself, through the sampled $\mathbf{z}_i$ and through the encoded counts of Equation (12); we abbreviate the posterior as $q(\mathbf{w}_i)$ where its arguments are clear. The divergence $\text{KL}(q \| p)$ reads as how far this cell deviates from its expected response, an anomaly score that the objective provides at no extra cost; whether it carries usable per-cell information at the operating point is an empirical question for the evaluation. The prior is also what would make counterfactuals possible, by drawing $\mathbf{w}' \sim p(\mathbf{w} | \mathbf{c}', t)$ and decoding, with abduction $\mathbf{w}'_i = \mathbf{w}_i + m_\psi(\mathbf{c}', t_i) - m_\psi(\mathbf{c}_i, t_i)$ carrying the cell's own residual unchanged: because the prior scale is fixed, the per-cell counterfactual is exactly a translation of the prior mean, and the difference $\mathbf{w}'_i - \mathbf{w}_i$ is manifestly free of the per-type gauge of Section 2.8. The type-level form of this machinery, in which the prior mean alone is moved to a new context and the cell's intrinsic state is held fixed, is exercised in the evaluation; the per-cell form with abduction is not, since it presupposes a non-degenerate response posterior.

The direct path from $\mathbf{x}_i$ into the posterior is necessary. The conditional independence $\mathbf{w} \perp \mathbf{x} | \mathbf{z}, \mathbf{c}$ is false by explaining-away: $\mathbf{x}$ is generated from both $\mathbf{z}$ and $\mathbf{w}$, so knowing $\mathbf{x}$ and $\mathbf{z}$ informs $\mathbf{w}$ directly, which is also why Slavutsky et al. (2026) condition their condition-specific posterior on the data rather than on the shared code. In our setting $\mathbf{z}$ is lossy in exactly the wrong direction: it is low-dimensional, the invariance regulariser removes niche-predictable variation from it, and that variation *is* the spatial response that $\mathbf{w}$ exists to capture. Without the direct path, $\mathbf{w}_i$ collapses to $\mathbb{E}[\text{response} | \text{type}, \text{niche}]$, identical for co-located similar cells, and the divergence vanishes (Appendix B.4).

Learned and fixed variances. Both posteriors carry learned diagonal variances; this is what makes the reparameterised sample meaningful (Kingma and Welling, 2014; Rezende et al., 2014) and lets uncertainty vary per cell, so that a 30-count cell can have a wider $q(\mathbf{z})$ than a 300-count cell. The prior scale σ_w is fixed at one, for two reasons of which the first is decisive. A learned *constant* σ_w is a gauge freedom: $\mathbf{w}$ and $\mathbf{B}$ enter the decoder only as $\mathbf{B}\mathbf{w}_i$, so rescaling dimension k of $\mathbf{w}$ by s and column k of $\mathbf{B}$ by $1/s$ leaves the likelihood unchanged, and a learned constant prior scale would absorb the compensating change in the prior; fixing it pins the scale. A learned *conditional* $\sigma_w(\mathbf{c}, t)$ would be meaningful, claiming that some niches produce more variable responses than others, but it opens a degenerate direction: the model can shrink $\text{KL}(q||p)$ for free by inflating $\sigma_w(\mathbf{c}, t)$ wherever the penalty bites, making the prior vague rather than making $\mathbf{w}$ predictable. If it is wanted, it must be bounded (Appendix B.7).

Decoder. The decoder is Equation (4): a network $a(\mathbf{z}_i) \in \mathbb{R}^G$ plus the linear shift $\mathbf{B}\mathbf{w}_i$, followed by a softmax over genes. It takes no type input, for the reason given in Section 2.2. If reconstruction is poor the remedy is to widen $\mathbf{z}$ or the network, not to add t .

2.4 Objective

With the neighbour quantities $\mathbf{c}_i$ and $\bar{\rho}_i$ held constant (Section 2.6), the evidence lower bound on $\log p(\mathbf{x}_i | \mathbf{c}_i, t_i, \bar{\rho}_i)$, conditional on the neighbours’ clean compositions (the coupled model is treated in Appendix A.1), under the variational family $q(\mathbf{z}_i) q(\mathbf{w}_i | \mathbf{c}_i, t_i, \mathbf{z}_i, \mathbf{x}_i)$ is

$$L_i = \mathbb{E}_{q(\mathbf{z}_i)} \left[\mathbb{E}_{q(\mathbf{w}_i | \mathbf{z}_i)} \left[\log p(\mathbf{x}_i | \mathbf{z}_i, \mathbf{w}_i) \right] - \text{KL} (q(\mathbf{w}_i | \mathbf{z}_i) \| p(\mathbf{w}_i | \mathbf{c}_i, t_i)) \right] - \text{KL} (q(\mathbf{z}_i) \| p(\mathbf{z}_i)), \quad (15)$$

where the $\mathbf{w}$ -divergence sits inside the outer expectation because $q(\mathbf{w}_i | \cdot)$ conditions on the sampled $\mathbf{z}_i$; evaluating the closed-form Gaussian divergence at the one reparameterised draw is an unbiased one-sample estimate of it (Appendix A.1). A second bound on the same evidence is obtained by restricting the variational family to $q(\mathbf{w}_i) := p(\mathbf{w}_i | \mathbf{c}_i, t_i)$, which zeroes that divergence:

$$L_i^z = \mathbb{E}_{q(\mathbf{z}_i)} \mathbb{E}_{p(\mathbf{w}_i | \mathbf{c}_i, t_i)} \left[\log p(\mathbf{x}_i | \mathbf{z}_i, \mathbf{w}_i) \right] - \text{KL} (q(\mathbf{z}_i) \| p(\mathbf{z}_i)). \quad (16)$$

We call this the *intrinsic path*, and it is load-bearing. The first reconstruction term does reward $\mathbf{z}$ for carrying signal, since $a(\mathbf{z})$ is the only route to all G genes, but with the counts also wired into $q(\mathbf{w})$ the allocation between $\mathbf{z}$ and $\mathbf{w}$ is ambiguous, and the second bound

breaks the tie toward $\mathbf{z}$: under it $\mathbf{z}$ must explain the cell given only the *expected* response. Because it runs through the same leakage mixture as the first bound, it never asserts $\kappa = 0$ and so never contradicts it (Appendix A.2).

The training objective sums the two bounds, with a weight ω on the intrinsic path and per-term weights on the divergences,

$$J = \frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} \left[\frac{1}{\bar{\ell}_i} \sum_g x_{ig} \log p_{ig}(\mathbf{z}_i, \mathbf{w}_i) + \omega \frac{1}{\bar{\ell}_i} \sum_g x_{ig} \log p_{ig}(\mathbf{z}_i, \check{\mathbf{w}}_i) - (1 + \omega) \alpha_z \text{KL} (q(\mathbf{z}_i) \| \mathcal{N}(\mathbf{0}, \mathbf{I})) - \alpha_w \text{KL} (q(\mathbf{w}_i) \| \mathcal{N}(m_\psi(\mathbf{c}_i, t_i), \sigma_w^2 \mathbf{I})) \right] - \alpha_a \text{Pen}, \quad (17)$$

where $\mathcal{S}$ is the set of cells whose loss is evaluated in a batch, $\check{\mathbf{w}}_i \sim p(\mathbf{w}_i | \mathbf{c}_i, t_i)$ is a draw from the prior, and Pen is the invariance term of Section 2.5, a functional of the whole batch; under the adversarial escalation it is replaced by the per-cell mean $\frac{1}{|\mathcal{S}|} \sum_i \text{Adv}_i$. One decoder is evaluated twice, with $\mathbf{w}$ from the posterior in the first reconstruction term and from the prior in the second. The factor $(1 + \omega)$ is what would keep J proportional to $L_i + \omega L_i^z$, at $\alpha_z = \alpha_w = 1/\bar{\ell}$ and $\alpha_a = 0$: each of Equations (15) and (16) carries its own $-\text{KL}(q(\mathbf{z}_i) \| p(\mathbf{z}_i))$, so their weighted sum contains $(1 + \omega)$ copies of it, as in Slavutsky et al. (2026, eq. 6), and omitting the second copy, the natural mistake, breaks the correspondence even at unit weights.

Scaling. The reconstruction terms are divided by ℓ_i . Unscaled, they grow in proportion to a cell’s depth and exceed the divergences by orders of magnitude, and depth differs widely between tissues, so weights tuned on one section would not transfer to another. The scaling changes what the weights mean: reconstruction becomes $O(1)$ while the divergences stay absolute, so $\alpha_z = 1$ prices a latent’s information roughly $\bar{\ell}$ -fold above the unscaled bound and collapses $\mathbf{z}$. At $\alpha_z = \alpha_w = 1/\bar{\ell}$, with $\bar{\ell}$ the section’s count scale, J is proportional to $L_i + \omega L_i^z$ for a cell of that depth, and that point, not 1, is the natural reference for the weights. We take $\bar{\ell}$ as the section’s median total count, with the primary section’s value fixed during development, and set $\alpha_z = \frac{1}{2}\bar{\ell}$. For the response the operating point departs from the reference deliberately: $\alpha_w = 0.1$, far above $1/\bar{\ell}$, because at lower weights the response begins to take over type structure from $\mathbf{z}$ and the agreement guard of Section 2.7 falls, most on the deepest section. At that weight the response posterior stays close to its prior mean, so the response readouts are in effect readouts of the context regression $m_\psi(\mathbf{c}_i, t_i)$;

what opening the per-cell channel would cost and buy is reported as a sensitivity analysis (Section 3.5). We use $\alpha_z = \frac{1}{2}/\bar{\ell}$, fixed on a ladder over $\{\frac{1}{4}, \frac{1}{2}, 1, 2\} \times 1/\bar{\ell}$ read on the diagnostics of Section 2.7: because both bounds carry the $\mathbf{z}$ -divergence, the objective charges it $(1 + \omega)$ -fold, and at $\omega = 1$ the half weight restores a single charge at the bound-equivalent point. A per-cell scaling against global weights also means that a low-count cell’s divergences are weighted more heavily relative to its likelihood than a high-count cell’s, pulling sparse cells toward the prior; we accept this as the price of transferable weights. Once any weight departs from $1/\bar{\ell}$, or through the per-cell scaling itself, J is a weighted surrogate rather than a bound; and because one $q(\mathbf{z})$ is shared by two variational families, q is a compromise between them, not the best posterior approximation under either, even at the bound-equivalent weights. The two reconstruction paths are bounds on the same evidence (Appendix A.2), but J , their scaled sum together with the invariance term, is a training criterion, and training does not maximise it as a single function: neighbour compositions enter under a stop-gradient, so each step treats the current $\bar{\rho}$ as data and ascends a partial gradient (Appendix A.1), and under the adversary the fit is a saddle point of a two-player game. We therefore claim no likelihood-based uncertainty for q ; the reported variability comes from the leakage sweep and the seeds. The hyperparameters are therefore ω , α_z , α_w and the response width d_w , plus α_a and, under escalation, the adversary’s update schedule; κ is not a hyperparameter but the sweep axis.

2.5 Conditional Invariance of $\mathbf{z}$

Architecture alone does not keep the environment out of $\mathbf{z}$. The counts $\mathbf{x}_i$ carry environmental signal, and with $d_z > d_w$ the intrinsic latent will absorb it given the chance. The target is

$$I(\mathbf{z}_i; [\mathbf{y}_i, \Phi_i] \mid t_i) = 0. \quad (18)$$

Two features of this target are deliberate. It is *conditional on type*, the equalised-odds form of an invariance constraint (Hardt et al., 2016; Zhang et al., 2018). The marginal constraint $I(\mathbf{z}; \mathbf{y}) = 0$ is the analogue of demographic parity, which must discard label information whenever label and protected variable are dependent (Zhao and Gordon, 2019). In tissue they are, because types cluster in space: carrying “I am a T cell” predicts “my neighbours are T cells”. A collapse of the code’s agreement with cell type under such an invariance, with the nuisance information driven to zero, is also documented for conditional subspace VAEs (Klys et al., 2018; Slavutsky et al., 2026). The cost of conditioning is that intrinsic variation that is spatially organised within a type is pushed into $\mathbf{w}$ to the extent that composition and image predict it, tumour subclones being

the sharp case; the decomposition is relative to t , not absolute, and lineage-level labels are preferable to finer or state-defined ones, for several reasons including a selection one (Appendix B.5). It also targets *both* $\mathbf{y}$ and Φ . If only composition were penalised, $\mathbf{z}$ would be free to absorb the morphology-defined niche (vessel proximity, fibrosis, density, necrosis), and $\mathbf{w}$ would under-report morphology-driven effects; $\mathbf{y}$ and Φ are correlated, so the part that matters is the residual that is image-predictable but not composition-predictable, and that residual is what would go unpenalised. For Φ , what the regulariser and the probe see is a summary of its leading principal components, so image information outside them, which the prior on $\mathbf{w}$ receives in full, is neither penalised nor probed. Neither the penalty below nor the adversary is part of any bound; both are posterior regularisation (Ganchev et al., 2010), a constraint on the variational family written as a penalty, which is the status the adversarial term also has in Slavutsky et al. (2026) (Appendix A.3).

Closed-form penalty. The first regulariser is the Gaussian estimate of the conditional mutual information, which needs no second optimiser. With $\mathbf{v}_i = [\mathbf{y}_i^{(-K)}, \text{PC}(\Phi_i)]$, where $\mathbf{y}_i^{(-K)}$ drops one composition column (the simplex constraint makes the full block singular) and $\text{PC}(\Phi)$ keeps the leading principal components of the image embedding,

$$\text{Pen} = \sum_t p(t) \cdot \frac{1}{2} \left[\log \det \hat{\Sigma}_{\mathbf{v}|t} + \log \det \hat{\Sigma}_{\mathbf{z}|t} - \log \det \hat{\Sigma}_{[\mathbf{z}, \mathbf{v}]|t} \right], \quad (19)$$

evaluated on the posterior means μ_z rather than on samples, since encoder noise would dilute the measured dependence and hiding dependence behind noise is exactly what the penalty must not permit (Appendix A.4). Three estimation details are not optional at realistic batch sizes. The image block is kept small (a dozen principal components), which keeps the joint block well conditioned and retains the components that carry the niche signal. The per-type moments are accumulated as exponential moving averages across batches (rate 0.05, an effective window of about twenty batches), because a spatial batch contains few cells of any rare type; a type enters the penalty only once its window-weighted cell count reaches ten times the joint dimension, and the fraction of cells excluded by this support floor is reported, since those cells go unregularised. The type weights $p(t)$ are the section-wide type frequencies and are not renormalised after an exclusion. Regularisation shrinks toward the diagonal (Ledoit and Wolf, 2004; Schäfer and Strimmer, 2005), $\hat{\Sigma} \leftarrow (1 - \gamma)\hat{\Sigma} + \gamma \text{diag}(\hat{\Sigma})$, which damps only the correlations; an additive ridge $+\lambda \mathbf{I}$ would inflate every variance by an amount that depends on how ill-conditioned

the estimate was, biasing the log-determinant upward most for rare types and weakening the penalty exactly where data are thinnest. Two further details concern the gradient. The value of Equation (19) is taken from the moving-average moments but its gradient from the current batch at full scale, a straight-through estimator (Bengio et al., 2013); otherwise the only differentiable path runs through the averaging weight, and the effective strength of α_a would silently scale with the averaging constant. And the log-determinants are computed on correlation rather than covariance matrices: mutual information is invariant to per-dimension rescaling, so the value is identical, while the gradient of a log-determinant, which involves the inverse matrix, stays bounded when a dimension has near-zero variance, as the composition of cells deep inside one niche does. A jitter of $10^{-5}\mathbf{I}$ is added to the unit-diagonal correlation matrix before the factorisation; on a correlation matrix this is a uniform relative perturbation, not the variance-dependent ridge rejected above.

Adversarial escalation. The Gaussian penalty removes linear dependence only. When the probe below finds that dependence survives it, the penalty is replaced by an adversary (Louizos et al., 2016; Xie et al., 2017; Ganin et al., 2016; Zhang et al., 2018): two heads with parameters $\xi, \hat{\mathbf{y}}_\xi, \hat{e}_\xi^\Phi : (\boldsymbol{\mu}_{z,i}, t_i) \mapsto \Delta^{K-1}, \Delta^{E_\Phi-1}$, trained by a separate optimiser to predict the composition and an image-niche membership from the posterior mean of the intrinsic code (evaluated on the mean for the same reason as Equation (19)), and an encoder term

$$\begin{aligned} \text{Adv}_i = \lambda_y \Big[& \text{CE}(\mathbf{y}_i, \bar{\mathbf{y}}(t_i)) - \text{CE}(\mathbf{y}_i, \hat{\mathbf{y}}_\xi(\boldsymbol{\mu}_{z,i}, t_i)) \Big] \\ & + \text{CE}(e_i^\Phi, \bar{\Phi}(t_i)) - \text{CE}(e_i^\Phi, \hat{e}_\xi^\Phi(\boldsymbol{\mu}_{z,i}, t_i)), \end{aligned} \quad (20)$$

which is the heads’ excess predictive skill over knowing the type alone, in categorical cross-entropy. The adversary maximises it, the encoder minimises it, and it is zero at the optimum; the two halves are reported separately. The image target $e_i^\Phi \in \Delta^{E_\Phi-1}$ is a soft cluster membership of Φ_i over E_Φ image niches, obtained by k -means on its leading principal components (never on the raw high-dimensional embedding, where distances are noise-dominated), fitted once and frozen. We set $E_\Phi = K$ so that the two heads face comparable difficulty, and weight the composition half by $\lambda_y = 3$, in the encoder term and in the heads’ own loss: composition is the channel of the leakage confound, and at equal weights the probe below still finds within-type composition information in $\mathbf{z}$ that a more heavily weighted composition head removes (Table 8); soft rather than hard assignments give a smoother target with no boundary artefacts. The adversary and the probe below therefore target the image through

different summaries, discrete memberships and continuous components respectively; the memberships are the natural target for a classification head, the components for the Gaussian penalty and its probe. A weak adversary is worse than none: heads that are updated too slowly are defeated by the encoder while an independent probe still finds the leak, so head strength must be verified by the probe, never by the heads’ own training loss. This is the observation of Elazar and Goldberg (2018) that adversarial removal is not enough without an independent check.

The probe, and how to decide. Whether the regulariser worked is decided by a *fresh* predictor of the invariance block $\mathbf{v}_i$ from $(\boldsymbol{\mu}_{z,i}, t_i)$, trained on a spatially separate training split and evaluated on held-out cells: never the training adversary, and never the training cells. We score it block by block, the composition block ($K-1$ components) and the image block (twelve leading principal components) separately, as the held-out gain in Gaussian log-likelihood per component over the type-only baseline, with each component’s variance fitted:

$$G_b = \frac{1}{|b|} \sum_{k \in b} \frac{1}{2} \log \frac{\mathbb{E}(v_k - \bar{v}_k(t))^2}{\mathbb{E}(v_k - \hat{v}_k(\boldsymbol{\mu}_z, t))^2}, \quad (21)$$

where $\hat{v}_k$ is the probe’s prediction of component k and $\bar{v}_k(t)$ its mean over cells of type t . G_b estimates the conditional predictive $\mathcal{V}$ -information from $\boldsymbol{\mu}_z$ to block b given the type, for the model family $\mathcal{V}$ of the probe (Xu et al., 2020), and $\exp(2G_b) - 1$ is the fraction of the block’s within-type variance that the probe explains. Because it is scale-free, each block counts in its own right: pooled over raw components, the variance of the image components would swamp the composition components, which carry the leakage confound. Two reference scales make G_b interpretable: the *noise floor*, its value when $\boldsymbol{\mu}_z$ is permuted across cells within type, and the *uncontrolled baseline*, its value for a model fit at $\alpha_a = 0$. Both are computed per block and per probe family, and G_b is reported as its excess over the floor, in nats per component, and as a fraction of the uncontrolled baseline’s excess. A probe at its floor bounds the dependence it can detect; it does not certify independence. Figure 10 illustrates the protocol. The probe is fitted in two families, a ridge regression and a small multilayer perceptron (Table 5), each against its own floor, and the decision is taken on the nonlinear one, since the Gaussian penalty can leave nonlinear dependence in place by construction. The rule is to escalate from Equation (19) to Equation (20) when the nonlinear probe of a converged closed-form fit is clearly above the floor, and then to take the smallest α_a that reduces the probe’s excess, on both blocks and for both probe families, to at most a quarter of the

uncontrolled baseline’s without a loss in the agreement between $\mathbf{z}$ and cell type or in held-out reconstruction. The agreement is the normalised mutual information (Strehl and Ghosh, 2002) between t and a k -means clustering of μ_z with $k = K$ clusters. If no such α_a exists, the invariance is costing cell type and the label set is too fine. **[TODO: the guard is not met on every section at the operating point (Table 8); the escalation itself was triggered once, on the primary section, and needs a closed-form fit per section to be shown]**

2.6 Frozen Neighbours and Batching

Neighbour quantities are treated as data, not as things to back-propagate through. The neighbour compositions ρ_j in Equation (5) enter under a stop-gradient. The mechanism this prevents is a degeneracy rather than an explosion: with the path open, cell j ’s clean composition is trained to absorb cell i ’s residual through $\bar{\rho}_i$, so ρ_j stops being j ’s own profile, and the coupling feeds back through $\bar{\rho}_j$. Ergen and Yosef (2025) disable gradients on precisely this path, for training stability; a first-order feedback argument bounds the amplification through the coupling by $1/(1-\kappa)$ (Appendix A.6). The neighbours’ responses $\mathbf{w}_j$ needed for ρ_j are posterior draws computed in the same batched pass and likewise placed under a stop-gradient.

A tempting shortcut is to compute the foreign influx from the baseline programme alone, $\rho_j \approx \text{softmax}(a(\mathbf{z}_j))$, which avoids the neighbours’ context entirely. It errs in the direction that matters most. Neighbours share niches, so $\mathbf{w}_j \approx \mathbf{w}_i$; dropping $\mathbf{w}_j$ strips the local spatial programme out of $\bar{\rho}_i$, and whatever the leakage term then fails to absorb is explained by $\mathbf{w}_i$ instead, *inflating* apparent spatial effects, the exact failure the leakage sweep exists to guard against.

Two hops, one pass. Because $\bar{\rho}_i$ needs ρ_j , hence $\mathbf{w}_j$, hence $\mathbf{c}_j$, hence $\mathcal{N}_j$, a training step for cell i reaches two graph hops even though the attention layer itself reaches one. Batching by *contiguous spatial tile* makes this cheap, because the halo of extra cells around a tile is shared rather than disjoint. A tile’s cells are the *seeds*; their graph neighbours form *ring one*, for which the context, the response posterior and the clean composition are all computed; the neighbours of ring one form *ring two*, for which only the type is looked up, since the attention sources are types. Both rings are built by look-ups in the same edge list that β uses, never by a distance query: a cell across a pruned edge is not in ring one even if a radius search would return it, and if the halo and the leak kernel disagree the foreign influx is silently wrong. Attention edges are restricted to destinations in the seeds and ring one,

since a ring-two cell’s context would normalise over a truncated neighbourhood. The loss is evaluated on seeds only (Figure 9). With tiles of a few thousand cells the halo adds a modest fraction to the number of nodes processed per step, falling with tile size, cheaper than a cache of clean compositions that must be kept current; scattered seeds, by contrast, would cost many times more nodes than a contiguous tile, so spacing seeds apart is the expensive option, not the cheap one. The arithmetic of a pass is dominated by the encoder and decoder layers, at $O(NGh)$ for hidden width h ; the foreign influx costs $O(|E|G)$ for $|E|$ directed edges, about $6NG$ at the Delaunay average degree, though its gathered edge-by-gene rows are the largest intermediate tensor of a step. One consequence of tile batching should be stated: the ring of a training tile can contain cells of a held-out tile, whose counts then enter training as inputs under a stop-gradient. No gradient flows from them and no loss is evaluated on them, but they are used transductively. The split fixes how much: 1–2% of training seeds have a held-out neighbour, and 5–10% of held-out cells enter some training step in this way, depending on the section; within two hops, where only a cell’s type is used, the share of seeds is 2–4%. We state the exposure rather than add a buffer, which would remove the same seeds.

2.7 Model Selection and Diagnostics

A model that explains everything with leakage reconstructs held-out cells well, so held-out reconstruction alone is not a valid stopping criterion. We stop on held-out reconstruction and on the agreement between $\mathbf{z}$ and cell type *jointly*. The agreement is the normalised mutual information between t and a k -means clustering of μ_z (Section 2.5), computed on a random subsample of all cells; the clustering is unsupervised by design, so the quantity asks whether the geometry of $\mathbf{z}$ still separates the types rather than whether a classifier could recover them. Because the encoder receives the label (Section 2.2), it measures whether $\mathbf{z}$ *retains* type, not whether it discovers it, and its collapse is the detector for the failure mode in which the invariance machinery empties $\mathbf{z}$ while the remaining reads still look well behaved. The weight on the response divergence is not applied at full strength from the first step: the coefficient multiplying $\text{KL}(q(\mathbf{w}_i) \| p(\mathbf{w}_i | \mathbf{c}_i, t_i))$ is $\alpha_w \min(1, e/30)$ at epoch e , so it rises linearly from zero and equals α_w from epoch 30 on; no other term is scheduled. Without this ramp the conditional prior can fail to open on a given seed, the posterior of $\mathbf{w}$ collapsing onto a prior that ignores $\mathbf{c}$ within the first ten epochs and never recovering; the collapse is detected as a divergence that stays at zero and as zero mutual information between $\mathbf{w}$ and the niche on held-out cells, and a fit that shows it is discarded and re-seeded. The

converged objective is unchanged. During the ramp the model is evaluated as usual but nothing is decided: no checkpoint is accepted, the patience count does not begin, and the running maximum of the type agreement is not updated. A checkpoint is accepted only if it improves held-out reconstruction while the agreement stands at or above a fixed fraction (0.9) of its running maximum over the fit. The maximum ratchets upward, so the bar rises as training finds a better $\mathbf{z}$, and a checkpoint that buys reconstruction at the cost of type structure is rejected outright rather than traded off. Held-out cells are entire spatial tiles, never random cells, because spatial autocorrelation leaks the answer through neighbouring cells landing on both sides of a random split (Roberts et al., 2017).

Budget and selection across seeds. Two budgets are used, and they are not interchangeable. Sweep points, where many fits must be comparable to each other rather than individually optimal, run a short budget (cosine annealing over 200 epochs, patience 20). Reference fits, the ones from which figures and single-fit analyses are taken, run a long budget (500 epochs, patience 40). The distinction matters because the objective is not unimodal in the seed: repeated fits of the same configuration at the long budget fall into more than one basin, and the better basin is not reached at the short budget at all. A reference model is therefore fitted from several random seeds and one member is selected, but never on held-out reconstruction alone: a fit can buy reconstruction while losing the intrinsic-state readouts, and at least one member of the family does. Selection is on the full diagnostic battery below, taken jointly — held-out reconstruction, the $\mathbf{z}$ -type agreement, the invariance probe, the mirror check, and the external-target readouts — with the remaining seeds reported as the envelope rather than discarded. Sweep trajectories are always read over their own seeds at the short budget and are not compared, point for point, with a long-budget reference fit.

Five diagnostics accompany every fit. (i) The within-type coefficient of determination of $\mathbf{z}_i$ regressed on $\mathbf{c}_i$, against a control in which $\mathbf{c}_i$ is permuted among cells of the same type; a pooled version would partly measure type separability, which both variables legitimately carry, and the within-type form is blind to it. This within-type check complements the mirror diagnostics of Appendix B.1; Section 2.3 closes both routes by which $\mathbf{c}_i$ could echo $\mathbf{z}_i$, and the check is retained as verification. (ii) $\text{KL}(q(\mathbf{w}_i) \| p(\mathbf{w}_i | \mathbf{c}_i, t_i))$ per dimension. Its collapse to zero has three causes that the training curves alone cannot distinguish: posterior collapse under a large α_w , the mirror, and a response posterior starved of ego evidence. Each has its own check and fix (Appendix B.4), and the mirror must be ruled out

before any free-bits style remedy (Kingma et al., 2016) is applied, since free bits would mask it and leave the anomaly score meaningless. (iii) The held-out probe of Equation (21). (iv) The $\mathbf{z}$ -type agreement, the hard floor whose collapse is the marginal-invariance failure mode. (v) Two degeneracy checks in the opposite direction, asking whether $\mathbf{z}$ has collapsed onto the label: the ratio $I(\mathbf{z}; t)/H(t)$ together with the within-type variance of μ_z , and the gap in held-out reconstruction between $\mathbf{z}$ and a one-hot type code [PENDING RUNS: compute for the canonical fits].

2.8 What Is Identified, and the Leakage Sweep

Identifiability. Four statements can be made. The leak fraction κ is not identified from the count matrix beyond an upper bound (Proposition 2), and is swept. Given κ , the response coordinates carry at least the symmetries of Proposition 1; we do not show that there are no others. The statement treats a , m_ψ and the posterior networks as arbitrary functions of their inputs, and each map is applied to every cell at once.

Proposition 1 (Symmetries of the response). *Fix κ , the graph and the kernel.*

- (a) Rotation. *For orthogonal $\mathbf{Q} \in \mathbb{R}^{d_w \times d_w}$, the map $\mathbf{w}_i \mapsto \mathbf{Q}\mathbf{w}_i$, $\mathbf{B} \mapsto \mathbf{B}\mathbf{Q}^\top$, $m_\psi \mapsto \mathbf{Q}m_\psi$ leaves the distribution of the counts in Equations (2) to (7) unchanged, and, with each posterior mapped to $\mathcal{N}(\mathbf{Q}\mu_{w,i}, \mathbf{Q}\text{diag}\nu_{w,i}^2\mathbf{Q}^\top)$, every term of the objective J of Equation (17). The mapped posterior is again diagonal, so J is unchanged within the family of Equation (14), when $\mathbf{Q}$ is a signed permutation.*
- (b) Constants over genes. *For any scalar function $c(\mathbf{z})$ and any $\mathbf{b} \in \mathbb{R}^{d_w}$, the map $a \mapsto a + c\mathbf{1}$, $\mathbf{B} \mapsto \mathbf{B} + \mathbf{1}\mathbf{b}^\top$ leaves every clean composition, and hence J , unchanged.*
- (c) Per-type translation. *Suppose there is a function τ with $\tau(\mathbf{z}) = t_i$ for every $\mathbf{z}$ in the support of $q(\mathbf{z}_i)$, for every cell i . For any $\delta_1, \dots, \delta_K \in \mathbb{R}^{d_w}$, the map $\mathbf{w}_i \mapsto \mathbf{w}_i + \delta_{t_i}$ (the posterior mean with it), $m_\psi(\mathbf{c}, t) \mapsto m_\psi(\mathbf{c}, t) + \delta_t$, $a(\mathbf{z}) \mapsto a(\mathbf{z}) - \mathbf{B}\delta_{\tau(\mathbf{z})}$ leaves J unchanged.*

A rescaling $\mathbf{w} \mapsto \mathbf{S}\mathbf{w}$, $\mathbf{B} \mapsto \mathbf{B}\mathbf{S}^{-1}$ with $\mathbf{S}$ diagonal and some $|S_{kk}| \neq 1$ is not a symmetry, because it changes the prior divergence. The centred shift $\mathbf{B}(\mathbf{w}_i - \bar{\mathbf{w}}_t)$, with $\bar{\mathbf{w}}_t$ the mean response of type t , and the realised prior shift $\mathbf{B}[m_\psi(\mathbf{c}_i, t_i) - m_\psi(\bar{\mathbf{c}}_t, t_i)]$, with $\bar{\mathbf{c}}_t$ its mean context, each centred over genes, are unchanged by all three maps.

The proof is in Appendix A.5. The diagonal posterior makes the rotation a symmetry of the model rather than of the fitted objective, and the fixed prior scale

is what removes the rescaling (Section 2.3). Loadings are read only after centring each column over genes. The per-type translation is the less obvious freedom. Its condition holds only approximately: the likelihood and the foreign influx pass through $a(\mathbf{z})$, which sees $\mathbf{z}$ but not t , and Gaussian posteriors of different types overlap, so the type can be read from $\mathbf{z}$ only to the extent that the $\mathbf{z}$ -type agreement (Section 2.7) keeps it readable. The translation is therefore a near-flat direction whose position the likelihood barely pins and the optimisation dynamics decide. Parameter weight decay does not fix it, because $\mathbf{w}$ is a network output and not a parameter, and penalising parameters only moves the offset further into $\mathbf{w}$. Two consequences are carried through the evaluation. The magnitude $\|\mathbf{w}_i\|$, and any ranking of types by it, is a statement about the gauge and not about how strongly a type responds to its context; the gauge-free quantities are the spread of $\mathbf{w}$ *within* a type, read through the centred shift $\mathbf{B}(\mathbf{w}_i - \bar{\mathbf{w}}_t)$ with $\bar{\mathbf{w}}_t$ the mean response of type t , and the realised prior shift $\mathbf{B}[m_\psi(\mathbf{c}_i, t_i) - m_\psi(\bar{\mathbf{c}}_t, t_i)]$ relative to $\bar{\mathbf{c}}_t$, the mean context of cells of type t . The two differ by the posterior residuals. And a “clean” per-cell profile must be decoded at such a reference context rather than at $\mathbf{w} = \mathbf{0}$, which would silently subtract the type offset along with the context effect. Differences between two contexts, which is what the counterfactual reads, are invariant to the gauge and need no correction. The allocation between $\mathbf{z}$ and $\mathbf{w}$ is not pinned by the likelihood, since both posteriors see the counts; it rests on the conditional-invariance constraint, on the intrinsic path, and on $d_w < d_z$. For $\mathbf{w}$ given the context this is in the spirit of the auxiliary-variable setting of Khemakhem et al. (2020), whose guarantee (recovery up to an affine map, under an injective additive-noise decoder and sufficient variability of the conditional prior) we do not establish for this likelihood; for $\mathbf{z}$ no comparable guarantee exists without the constraint (Locatello et al., 2019), and the constraint is what we verify with the probe rather than assume.

The sweep. The unit of inference is a one-parameter sensitivity analysis with κ as the sensitivity parameter (Rosenbaum, 2002): at each κ we report the fitted readout and its interval, not the set of all fits consistent with the data. Because the data bound κ from above but not from below (Proposition 2), the leak fractions compatible with the data form an interval from zero, an identified set in the sense of partial identification (Manski, 2003). The model is fitted independently at each $\kappa \in \{0, 0.05, 0.1, 0.2, 0.3, 0.4\}$ with several random seeds per grid point, with every other weight held at values calibrated once at a reference grid point and the held-out probe checked at every κ so that a change

in a readout is not a change in the invariance. Every readout of interest is reported as a trajectory over κ , with an interval at each grid point that pools the seeds with a spatial block bootstrap over tiles, so that it reflects sampling as well as optimisation: the within-type shifts of the response defined above, the spatial autocorrelation of each latent dimension (read within a fit), the held-out probes and the agreement with external labels. Findings are read through one quantity.

Definition 1 (Breakdown point). For a readout r with null value r_0 , let $I_\kappa(r)$ be its interval at leak fraction κ . The *breakdown point* $\kappa^*(r)$ is the smallest grid value at which $I_\kappa(r)$ contains r_0 or the sign of $r - r_0$ differs from its sign at $\kappa = 0$. If there is none, $\kappa^*(r)$ is reported as exceeding the top of the grid. The null value is zero for signed contrasts and, for magnitudes and for the spatial autocorrelation of the response, the value under a within-type permutation of the cells’ positions [TODO: specify the null per readout; a magnitude’s null may need a refit with permuted contexts].

Figure 8 illustrates the definition on schematic readouts. Rankings are read through pairwise differences, each with its own breakdown point. The breakdown point is reported rather than thresholded, so no verdict depends on where the grid stops. No readout is designated primary in advance: the final sweeps were preceded by development sweeps, so any such set would be chosen with knowledge of earlier results. Every readout is reported with its breakdown point, and its intervals are Bonferroni-adjusted over all readouts of the table that reports it. Findings are section-level. Where two serial sections of the same tissue are available, a tissue-specific finding is claimed only if it holds on both, and a property of the method, such as whether a readout moves with κ at all, is required on every section. A breakdown point above the grid rules out one specific confound: leakage of the modelled form, a section-wide fraction mixed through a fixed one-hop kernel. It does not rule out leakage that this form cannot represent (longer-ranged, gene-specific, or dependent on donor or receiver depth), which does not change with κ and would pass every grid point unchanged. A readout that survives the grid is therefore a finding relative to this confound, not a proof of biology; one that breaks down at the first grid point above zero is not separable even from leakage of the modelled form, and one whose interval contains its null already at $\kappa = 0$ is no finding at all. The sweep separates spatial association from leakage of that form. It does not separate a response from intrinsic variation that is spatially organised within a type, which conditional invariance assigns to $\mathbf{w}$, nor from cells selecting their niche. Some readouts are expected to move with κ by construction: as κ grows, more of the resemblance

between neighbours is assigned to the mixture and the within-type spread of the response shrinks, so a decline along the grid is mechanical, whereas a change in the *direction* of a programme or in which latent a target aligns with is not. Two consequences follow. Gene programmes are identified only up to the rotation and the per-column constants above, so raw loadings and probe directions are never compared across runs or seeds; only run-internal, rotation-invariant summaries are reported, and any gene-level statement is treated as relative to the run that produced it. And the two kinds of uncertainty are kept apart: the interval at a grid point is a sampling statement and narrows with more cells, whereas the trajectory across the grid is not a confidence interval and does not narrow, because the confound it expresses is not a sampling problem. An operating point on the grid is used for figures and for the analyses that require a single fit, and is always reported alongside the sweep.

What $\mathbf{w}$ is not. The context $\mathbf{c}_i$ is the environment: deterministic, shared by co-located cells, a conditioning variable. If a niche descriptor were all that was wanted, one would drop the latent and feed $\mathbf{c}_i$ to the decoder directly, which is roughly the node-centric model of Fischer et al. (2023) and is simpler. The response $\mathbf{w}_i$ is latent and per cell, and it earns its place because the per-cell direct path (Section 2.3) lets $\mathbf{B}$ and m_ψ be fitted against each cell’s own residual rather than a niche-level average; per-cell anomaly scores, per-cell responses and counterfactuals with abduction are the further uses it is designed for, and they are future work. At an α_w far above the bound-equivalent value the objective itself suppresses the divergence, so a small divergence shows how the objective is weighted, not that responses within a niche are homogeneous.

3 EXPERIMENTS

3.1 Setup

Sections. We fit DISCELL to four sections profiled with the same imaging-based assay and gene panel of about five thousand genes (Janesick et al., 2023): an ovarian cancer section preserved by formalin fixation and paraffin embedding (FFPE), our primary section (10x Genomics, 2024a); a lung cancer section, also FFPE (10x Genomics, 2024c); a fresh-frozen ovarian cancer section, sequenced about eight times deeper (10x Genomics, 2024b); and a core of a paediatric lung tissue microarray (TMA) from GEO series GSE315411 (Williams-Katek et al., 2026). A serial section of the same TMA core on a second slide is the held-out section: no model is fitted to it, and it is read only through the model fitted on its neighbour. The sections range

from about seventy thousand to over a million cells (Table 9).

Labels. Every section carries lineage-level labels, in which states and locations are merged into their lineage (Section 2.1). On the ovarian FFPE section and the TMA they are derived from curated annotations, and on the other two from marker-annotated clusters; the mappings were fixed before any fit. Cells without a label form one class, which enters training like any other and is never a target of evaluation.

Graph, kernel and image. The graph is Delaunay adjacency pruned at $40\ \mu\text{m}$, the leakage kernel is β with $\tau = 20\ \mu\text{m}$, and the image descriptor is the KRONOS embedding of the four-channel $128\ \mu\text{m}$ field with the $25\ \mu\text{m}$ disc over the cell set to zero, on every section (Section 2.1, Appendices B.2 and B.3).

Configuration. One configuration is used on every section (Table 5): a KL warm-up of the response over its first thirty epochs, the adversary with its composition term weighted three times its image term, in the heads’ loss and in the encoder’s, $\kappa = 0.1$ as the operating point, and a budget of 500 epochs with patience 40. Only α_z changes between sections, because it is set to $\frac{1}{2} \cdot 1/\bar{\ell}$ with $\bar{\ell}$ the section’s count scale: its median total count, with the primary section’s value fixed during development. The operating point $\kappa = 0.1$ is where the analyses that need a single fit are read, not an estimate of the leak fraction. It was taken before the full sweep, inside the published range of misassignment for the platform [NEEDS A SOURCE], and kept after it, because on the primary section it lies at the end of the plateau of held-out reconstruction, where the loadings agree best across seeds and the mirror diagnostic (Section 2.7) is already below its level at $\kappa = 0$. Held-out reconstruction scores the fit as an autoencoder and favours small κ , so it describes the point rather than selecting it, and no claim rests on the choice: every readout is reported across the sweep (Section 2.8).

Splits, seeds and read-outs. Each section is cut into contiguous tiles of at most 4,096 cells (2,048 on the TMA core), and 15% of the tiles are held out (Appendix C.3). Every configuration is fitted from three random seeds, and every fit is read at its accepted checkpoint. A readout is reported as the mean over seeds with its range, and, where a sampling interval is needed, with a spatial block bootstrap over tiles (Definition 1). The leakage sweep refits the same configuration at every $\kappa \in \{0, 0.05, 0.1, 0.2, 0.3, 0.4\}$ with three seeds each.

Baselines. We compare with resolVI (Ergen and Yosef, 2025), SIMVI (Dong et al., 2025), MintFlow (Akbarnejad et al., 2025) and Cellina (Moeed et al., 2026), each at its published defaults and on our graph wherever the method accepts an external one (Table 12). All methods see the same cells, split and labels, and are scored by the same read-outs on their intrinsic latent.

3.2 Model Quality Across Four Tissues

3.3 What the Intrinsic State Carries

3.4 What the Response Carries

3.5 The Leakage Sweep

3.6 Comparison with Other Models

4 CONCLUSION

DISCELL writes a cell’s counts as an intrinsic state, a response to its microenvironment and leakage from its neighbours, and makes explicit what the data leave open between the last two. The intrinsic state is kept free of the niche given the cell’s type by architecture and by a conditional invariance that a held-out probe grades independently of training. The leakage enters through a fixed kernel on the section’s own graph. The leak fraction is not estimated, since the data bound it only from above, but swept, and every spatial readout is reported with the leak fraction at which it breaks down.

Limitations. Each is argued where it arises; they are collected here.

- *The form of the leakage.* The leak fraction is one number per section, the leakage is the same for every gene and scales with the receiving cell’s depth, and there is no ambient term (Section 2.2). A per-cell or per-gene variation in contamination is not covered by the sweep; its sensitivity to the leak’s form is checked on the primary section only.
- *What the sweep separates.* It separates a spatial association from leakage of the modelled form, not from spatially organised intrinsic variation within a type, which the conditional invariance assigns to the response, nor from cells selecting their niche (Section 2.8).
- *Labels.* The labels come from the same contaminated counts, and the decomposition is relative to their granularity: a state merged into its lineage becomes variation the model must place, and a spatially clustered state merged in this way is read as a response (Sections 2.1 and 2.5).

- *The invariance is incomplete.* The adversary removes most, not all, of the neighbour composition a nonlinear probe can find in the intrinsic state, and a probe at its floor bounds only the dependence it can detect (Table 8).
- *The response at the operating point.* With the response weight far above its bound-equivalent value, the response readouts are in effect readouts of the context regression; the per-cell channel is closed by design (Section 2.4).
- *Inference.* The intrinsic encoder does not see the foreign influx, so its posterior mean can differ from the exact one, by an amount that is small in simulation and cannot be measured on tissue (Appendix D.6), and the posterior family is a compromise, so the uncertainty reported comes from the sweep and the seeds, not from the posterior (Section 2.3).
- *Identification.* Beyond the upper bound on the leak fraction and the invariance that is verified, nothing guarantees that the intrinsic and response coordinates are identified (Section 2.8).
- *Evaluation.* Cells of held-out tiles enter training as neighbours under a stop-gradient (Section 2.6), every fit is to one section, and only one section is held out whole. Two of the comparison methods could be run on only some of the sections (Table 13).

Future work. Three extensions follow directly. A perturbed and an unperturbed condition would add a third source of variation, so that the decomposition becomes intrinsic state, microenvironmental response and perturbation response, with the leakage sweep applied to all three. An operating point with an open per-cell response channel would support per-cell anomaly scores and counterfactuals with abduction. And a nuclear/extranuclear split of the transcripts would supply the gene a cell cannot express itself that the upper bound needs to become an estimate of the leak fraction (Proposition 2).

References

- 10x Genomics (2024a). FFPE human ovarian cancer with 5K human pan tissue and pathways panel plus 100 custom genes. <https://www.10xgenomics.com/datasets/xenium-prime-ffpe-human-ovarian-cancer>. In Situ Gene Expression dataset by Xenium Onboard Analysis v3.0.0, 10x Genomics (2024, December 17).
- 10x Genomics (2024b). Fresh frozen human ovarian adenocarcinoma with 5K human

- pan tissue and pathways panel. <https://www.10xgenomics.com/datasets/xenium-prime-fresh-frozen-human-ovary>. In Situ Gene Expression dataset by Xenium Onboard Analysis v3.0.0, 10x Genomics (2024, September 4).
- 10x Genomics (2024c). Post-Xenium technical note: Xenium v1 and Xenium Prime 5K for FFPE human lung cancer. <https://www.10xgenomics.com/datasets/xenium-human-lung-cancer-post-xenium-technote>. Experiment 2, Xenium Prime 5K. In Situ Gene Expression dataset by Xenium Onboard Analysis v3.0.0, 10x Genomics (2024, November 6).
- Akbarnejad, A., Steele, L., Jafree, D. J., Birk, S., Sallese, M. R., et al. (2025). Mapping and reprogramming human tissue microenvironments with MintFlow. *bioRxiv*.
- Arnol, D., Schapiro, D., Bodenmiller, B., Saez-Rodriguez, J., and Stegle, O. (2019). Modeling Cell-Cell Interactions from Spatial Molecular Data with Spatial Variance Component Analysis. *Cell Reports*, 29(1):202–211.e6.
- Bengio, Y., Léonard, N., and Courville, A. (2013). Estimating or propagating gradients through stochastic neurons for conditional computation. *arXiv preprint arXiv:1308.3432*.
- Bilous, M., Buszta, D., Bac, J., Kang, S., Dong, Y., Tissot, S., Andre, S., Alexandre Gaveta, M., Voize, C., Peters, S., Homicsko, K., and Gottardo, R. (2026). Resolving sensitivity, specificity and signal contamination in Xenium spatial transcriptomics. *Nature Methods*, 23:1152–1162.
- Birk, S., Lotfollahi, M., et al. (2025). Quantitative characterization of cell niches in spatially resolved omics data. *Nature Genetics*. CHECK: author list, volume, identifier.
- Brody, S., Alon, U., and Yahav, E. (2022). How attentive are graph attention networks? In *International Conference on Learning Representations*.
- Cable, D. M., Murray, E., Shanmugam, V., Zhang, S., Zou, L. S., Diao, M., Chen, H., Macosko, E. Z., Irizarry, R. A., and Chen, F. (2022). Cell type-specific inference of differential expression in spatial transcriptomics. *Nature Methods*, 19(9):1076–1087. Preprint: doi:10.1101/2021.12.26.474183.
- Cover, T. M. and Thomas, J. A. (2006). *Elements of Information Theory*. Wiley, 2nd edition.
- Dong, M., Kluger, Y., et al. (2025). SIMVI disentangles intrinsic and spatial-induced cellular states in spatial omics data. *Nature Communications*. CHECK: author list, volume, identifier.
- Elazar, Y. and Goldberg, Y. (2018). Adversarial removal of demographic attributes from text data. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 11–21.
- Ergen, C. and Yosef, N. (2025). ResolVI – addressing noise and bias in spatial transcriptomics. *bioRxiv* preprint.
- Fischer, D. S., Schaar, A. C., and Theis, F. J. (2023). Modeling intercellular communication in tissues using spatial graphs of cells. *Nature Biotechnology*, 41:332–336.
- Fleming, S. J., Chaffin, M. D., Arduini, A., Akkad, A.-D., Banks, E., Marioni, J. C., Philippakis, A. A., Ellinor, P. T., and Babadi, M. (2023). Unsupervised removal of systematic background noise from droplet-based single-cell experiments using CellBender. *Nature Methods*, 20(9):1323–1335. Preprint: doi:10.1101/791699.
- Ganchev, K., Graça, J., Gillenwater, J., and Taskar, B. (2010). Posterior regularization for structured latent variable models. *Journal of Machine Learning Research*, 11:2001–2049.
- Ganin, Y., Ustinova, E., Ajakan, H., Germain, P., Larochelle, H., Laviolette, F., Marchand, M., and Lempitsky, V. (2016). Domain-adversarial training of neural networks. *Journal of Machine Learning Research*, 17(59):1–35.
- Hardt, M., Price, E., and Srebro, N. (2016). Equality of opportunity in supervised learning. In *Advances in Neural Information Processing Systems*, volume 29.
- Hernán, M. A., Hernández-Díaz, S., and Robins, J. M. (2004). A structural approach to selection bias. *Epidemiology*, 15(5):615–625.
- Janesick, A., Shelansky, R., Gottscho, A. D., Wagner, F., Williams, S. R., Rouault, M., Beliakoff, G., Morrison, C. A., Oliveira, M. F., Sicherman, J. T., et al. (2023). High resolution mapping of the tumor microenvironment using integrated single-cell, spatial and in situ analysis. *Nature Communications*, 14:8353. CHECK: full author list truncated.
- Khemakhem, I., Kingma, D. P., Monti, R. P., and Hyvärinen, A. (2020). Variational autoencoders and nonlinear ICA: A unifying framework. In *International Conference on Artificial Intelligence and Statistics*.
- Kingma, D. P. and Ba, J. (2015). Adam: A method for stochastic optimization. In *International Conference on Learning Representations*.
- Kingma, D. P., Salimans, T., Jozefowicz, R., Chen, X., Sutskever, I., and Welling, M. (2016). Improved variational inference with inverse autoregressive flow. In

- Advances in Neural Information Processing Systems*, volume 29.
- Kingma, D. P. and Welling, M. (2014). Auto-encoding variational Bayes. In *International Conference on Learning Representations*.
- Klys, J., Snell, J., and Zemel, R. (2018). Learning latent subspaces in variational autoencoders. In *Advances in Neural Information Processing Systems*, volume 31.
- Ledoit, O. and Wolf, M. (2004). A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):365–411.
- Li, Y., Tian, X., Gong, M., Liu, Y., Liu, T., Zhang, K., and Tao, D. (2018). Deep domain generalization via conditional invariant adversarial networks. In *European Conference on Computer Vision*, pages 624–639.
- Locatello, F., Bauer, S., Lucic, M., Rätsch, G., Gelly, S., Schölkopf, B., and Bachem, O. (2019). Challenging common assumptions in the unsupervised learning of disentangled representations. In *International Conference on Machine Learning*.
- Loshchilov, I. and Hutter, F. (2017). SGDR: Stochastic gradient descent with warm restarts. In *International Conference on Learning Representations*.
- Louizos, C., Swersky, K., Li, Y., Welling, M., and Zemel, R. (2016). The variational fair autoencoder. In *International Conference on Learning Representations*.
- Madras, D., Creager, E., Pitassi, T., and Zemel, R. (2018). Learning adversarially fair and transferable representations. In *International Conference on Machine Learning*, pages 3384–3393.
- Manski, C. F. (2003). *Partial Identification of Probability Distributions*. Springer.
- Megas, S., Chen, D. G., Polanski, K., Asadollahzadeh, H., Eliasof, M., Schönlieb, C.-B., and Teichmann, S. A. (2026). Celcomen: spatial causal disentanglement for single-cell and tissue perturbation modeling. *Nature Communications*, 17:4126.
- Mitchel, J., Gao, T., Petukhov, V., Cole, E., and Kharchenko, P. V. (2026). Impact and correction of segmentation errors in spatial transcriptomics. *Nature Genetics*, 58:434–444.
- Moeed, A., Schrod, S., Rohbeck, M., Bonder, M. J., Lutsik, P., Stegle, O., and Dimitrov, D. (2026). Querying counterfactuals on tissue graphs with supervised disentanglement. arXiv preprint 2606.08493.
- Ni, Z., Prasad, A., Chen, S., Halberg, R. B., Arkin, L. M., Drolet, B. A., Newton, M. A., and Kendzior, C. (2022). SpotClean adjusts for spot swapping in spatial transcriptomics data. *Nature Communications*, 13:2971.
- Petukhov, V., Xu, R. J., Soldatov, R. A., Cadinu, P., Khodosevich, K., Moffitt, J. R., and Kharchenko, P. V. (2022). Cell segmentation in imaging-based spatial transcriptomics. *Nature Biotechnology*, 40:345–354.
- Pogodin, R., Deka, N., Li, Y., Sutherland, D. J., Veitch, V., and Gretton, A. (2023). Efficient conditionally invariant representation learning. In *International Conference on Learning Representations*.
- Rezende, D. J., Mohamed, S., and Wierstra, D. (2014). Stochastic backpropagation and approximate inference in deep generative models. In *International Conference on Machine Learning*.
- Roberts, D. R., Bahn, V., Ciuti, S., Boyce, M. S., Elith, J., Guillera-Arroita, G., Hauenstein, S., Lahoz-Monfort, J. J., Schröder, B., Thuiller, W., Warton, D. I., Wintle, B. A., Hartig, F., and Dormann, C. F. (2017). Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, 40(8):913–929.
- Rosenbaum, P. R. (2002). *Observational Studies*. Springer, 2nd edition.
- Schäfer, J. and Strimmer, K. (2005). A shrinkage approach to large-scale covariance matrix estimation and implications for functional genomics. *Statistical Applications in Genetics and Molecular Biology*, 4(1).
- Shaban, M., Chang, Y., Qiu, H., Yeo, Y. Y., Song, A. H., Jaume, G., et al. (2025). A foundation model for spatial proteomics. arXiv preprint 2506.03373.
- Slavutsky, Y., Beker, O., Blei, D., and Dumitrescu, B. (2026). Variational learning of disentangled representations. In *International Conference on Machine Learning*. arXiv:2506.17182.
- Strehl, A. and Ghosh, J. (2002). Cluster ensembles — a knowledge reuse framework for combining multiple partitions. *Journal of Machine Learning Research*, 3:583–617.
- Svensson, V., Teichmann, S. A., and Stegle, O. (2018). SpatialDE: identification of spatially variable genes. *Nature Methods*, 15:343–346.
- Tanevski, J., Ramirez Flores, R. O., Gabor, A., Schapiro, D., and Saez-Rodriguez, J. (2022). Explainable multiview framework for dissecting spatial relationships from highly multiplexed data. *Genome Biology*, 23:97.
- Tomczak, J. M. and Welling, M. (2018). VAE with a VampPrior. In *International Conference on Artificial Intelligence and Statistics*.

- Townes, F. W. and Engelhardt, B. E. (2023). Nonnegative spatial factorization applied to spatial genomics. *Nature Methods*, 20(2):229–238.
- Velten, B., Braunger, J. M., Argelaguet, R., Arnol, D., Wirbel, J., Bredikhin, D., Zeller, G., and Stegle, O. (2022). Identifying temporal and spatial patterns of variation from multimodal data using MEFISTO. *Nature Methods*, 19(2):179–186.
- Williams-Katek, A. L., Mallapragada, S., Mee, E. D., Fischer, B. K., Vannan, A., Eldredge, L. C., Deutsch, G. H., Kropski, J. A., Sucre, J. M., and Banovich, N. E. (2026). Fishing with two lines: a hybrid approach to spatial transcriptomics discovery. *Life Science Alliance*, 9(10).
- Xie, Q., Dai, Z., Du, Y., Hovy, E., and Neubig, G. (2017). Controllable invariance through adversarial feature learning. In *Advances in Neural Information Processing Systems*, volume 30.
- Xu, Y., Zhao, S., Song, J., Stewart, R., and Ermon, S. (2020). A theory of usable information under computational constraints. In *International Conference on Learning Representations*.
- Yang, S., Corbett, S. E., Koga, Y., Wang, Z., Johnson, W. E., Yajima, M., and Campbell, J. D. (2020). Decontamination of ambient RNA in single-cell RNA-seq with DecontX. *Genome Biology*, 21:57. Preprint: doi:10.1101/704015.
- Young, M. D. and Behjati, S. (2020). SoupX removes ambient RNA contamination from droplet-based single-cell RNA sequencing data. *GigaScience*, 9(12):giaa151.
- Zhang, B. H., Lemoine, B., and Mitchell, M. (2018). Mitigating unwanted biases with adversarial learning. In *Proceedings of the 2018 AAAI/ACM Conference on AI, Ethics, and Society*, pages 335–340.
- Zhao, H. and Gordon, G. J. (2019). Inherent tradeoffs in learning fair representations. In *Advances in Neural Information Processing Systems*, volume 32.

Checklist

[TODO: Answer at submission time. Questions must not be modified; the instruction paragraph of the template has been removed.]

1. For all models and algorithms presented, check if you include:
 - (a) A clear description of the mathematical setting, assumptions, algorithm, and/or model. [Yes]

- (b) An analysis of the properties and complexity (time, space, sample size) of any algorithm. [Yes]
 - (c) (Optional) Anonymized source code, with specification of all dependencies, including external libraries. [No]
2. For any theoretical claim, check if you include:
 - (a) Statements of the full set of assumptions of all theoretical results. [Yes]
 - (b) Complete proofs of all theoretical results. [Yes]
 - (c) Clear explanations of any assumptions. [Yes]
3. For all figures and tables that present empirical results, check if you include:
 - (a) The code, data, and instructions needed to reproduce the main experimental results (either in the supplemental material or as a URL). [No]
 - (b) All the training details (e.g., data splits, hyperparameters, how they were chosen). [Yes]
 - (c) A clear definition of the specific measure or statistics and error bars (e.g., with respect to the random seed after running experiments multiple times). [Yes]
 - (d) A description of the computing infrastructure used. (e.g., type of GPUs, internal cluster, or cloud provider). [Yes]
4. If you are using existing assets (e.g., code, data, models) or curating/releasing new assets, check if you include:
 - (a) Citations of the creator If your work uses existing assets. [Yes]
 - (b) The license information of the assets, if applicable. [Yes]
 - (c) New assets either in the supplemental material or as a URL, if applicable. [Not Applicable]
 - (d) Information about consent from data providers/curators. [Not Applicable]
 - (e) Discussion of sensible content if applicable, e.g., personally identifiable information or offensive content. [Not Applicable]
5. If you used crowdsourcing or conducted research with human subjects, check if you include:
 - (a) The full text of instructions given to participants and screenshots. [Not Applicable]
 - (b) Descriptions of potential participant risks, with links to Institutional Review Board (IRB) approvals if applicable. [Not Applicable]

- (c) The estimated hourly wage paid to participants and the total amount spent on participant compensation. [Not Applicable]

DISCELL: Supplementary Materials

A DERIVATIONS

A.1 The Reconstruction Bound

Fix a cell i and hold $\mathbf{c}_i$ and $\bar{\rho}_i$ constant, as Section 2.6 does; every probability in this appendix is conditional on $\bar{\rho}_i$, which we suppress from the notation. Then

$$\log p(\mathbf{x}_i | \mathbf{c}_i, t_i) = \log \int \int p(\mathbf{x}_i | \mathbf{z}, \mathbf{w}) p(\mathbf{w} | \mathbf{c}_i, t_i) p(\mathbf{z}) d\mathbf{z} d\mathbf{w} \quad (22)$$

$$\geq \mathbb{E}_{q(\mathbf{z})} \mathbb{E}_{q(\mathbf{w}|\mathbf{z})} \left[\log p(\mathbf{x}_i | \mathbf{z}, \mathbf{w}) + \log p(\mathbf{w} | \mathbf{c}_i, t_i) + \log p(\mathbf{z}) - \log q(\mathbf{z}) - \log q(\mathbf{w} | \mathbf{z}) \right] \quad (23)$$

$$= \mathbb{E}_{q(\mathbf{z})} \left[\mathbb{E}_{q(\mathbf{w}|\mathbf{z})} \left[\log p(\mathbf{x}_i | \mathbf{z}, \mathbf{w}) \right] - \text{KL}(q(\mathbf{w} | \mathbf{z}) \| p(\mathbf{w} | \mathbf{c}_i, t_i)) \right] - \text{KL}(q(\mathbf{z}) \| p(\mathbf{z})), \quad (24)$$

where Equation (23) is Jensen's inequality under the variational family $q(\mathbf{z}) q(\mathbf{w} | \mathbf{c}_i, t_i, \mathbf{z}, \mathbf{x}_i)$. Because $q(\mathbf{w} | \cdot)$ conditions on the sampled $\mathbf{z}$, the exact bound keeps the $\mathbf{w}$ -divergence inside $\mathbb{E}_{q(\mathbf{z})}$. The divergence between two diagonal Gaussians is available in closed form,

$$\text{KL}(\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\sigma}^2) \| \mathcal{N}(\mathbf{m}, \mathbf{s}^2)) = \sum_k \left[\log \frac{s_k}{\sigma_k} + \frac{\sigma_k^2 + (\mu_k - m_k)^2}{2s_k^2} - \frac{1}{2} \right], \quad (25)$$

and evaluating it at the single reparameterised draw of $\mathbf{z}$ used for the reconstruction term is an unbiased one-sample estimate of $\mathbb{E}_{q(\mathbf{z})}[\text{KL}(\cdot)]$.

Summed over cells, with each neighbour's $\mathbf{z}_j, \mathbf{w}_j$ drawn from its own posterior as in Section 2.6, $\sum_i L_i$ is a one-sample estimate of the evidence lower bound of the coupled model, $\log p(\{\mathbf{x}_i\} | \{\mathbf{c}_i, t_i\})$, under the fully factorised family $\prod_i q(\mathbf{z}_i) q(\mathbf{w}_i | \mathbf{z}_i)$. The per-cell bounds are not independent terms, since $\bar{\rho}_i$ is a function of the neighbours' latents. The stop-gradient removes the terms $i \neq j$ from the gradient with respect to q_j , so training ascends a partial gradient of this bound: a fixed-point iteration in which each step treats the current $\bar{\rho}$ as data.

A.2 The Intrinsic Path and the Factor $(1 + \omega)$

A second bound on the same evidence holds by restricting the variational family to $q(\mathbf{w}) := p(\mathbf{w} | \mathbf{c}_i, t_i)$, which zeroes the $\mathbf{w}$ -divergence and leaves

$$\log p(\mathbf{x}_i | \mathbf{c}_i, t_i) \geq \mathbb{E}_{q(\mathbf{z})} \mathbb{E}_{p(\mathbf{w}|\mathbf{c}_i, t_i)} \left[\log p(\mathbf{x}_i | \mathbf{z}, \mathbf{w}) \right] - \text{KL}(q(\mathbf{z}) \| p(\mathbf{z})) =: L_i^z. \quad (26)$$

Call the bound of Appendix A.1 L_i . Each carries its own $-\text{KL}(q(\mathbf{z}_i) \| p(\mathbf{z}_i))$, so $L_i + \omega L_i^z$ contains $-(1 + \omega) \text{KL}(q(\mathbf{z}_i) \| p(\mathbf{z}_i))$. Omitting the second copy breaks the correspondence with $L_i + \omega L_i^z$ even at $\omega = 1$.

The construction mirrors the two-bound objective of Slavutsky et al. (2026), where the sum of the shared-only and joint bounds corresponds to minimising $\text{KL}(q(\mathbf{z}) \| p(\mathbf{z} | \mathbf{x})) + \text{KL}(q(\mathbf{z})q(\mathbf{w}) \| p(\mathbf{z}, \mathbf{w} | \mathbf{x}, u))$ with u their condition label, a sum of two non-negative divergences. Here, unlike in Slavutsky et al. (2026), whose shared-only term decodes from $\mathbf{z}$ alone and so bounds a different model, both terms bound the same evidence,

$$L_i + \omega L_i^z \leq (1 + \omega) \log p(\mathbf{x}_i | \mathbf{c}_i, t_i),$$

with gap $\text{KL}(q(\mathbf{z})q(\mathbf{w} | \mathbf{z}) \| p(\mathbf{z}, \mathbf{w} | \mathbf{x}_i, \mathbf{c}_i, t_i)) + \omega \text{KL}(q(\mathbf{z})p(\mathbf{w} | \mathbf{c}_i, t_i) \| p(\mathbf{z}, \mathbf{w} | \mathbf{x}_i, \mathbf{c}_i, t_i))$. One $q(\mathbf{z})$ serves two variational families, so its optimum is a compromise and not the best approximation of the posterior under either, which is one of the reasons uncertainty is reported from the leakage sweep rather than from q . The invariance term sits outside the bound for a separate reason (Appendix A.3).

Why the intrinsic path is load-bearing: in Slavutsky et al. (2026) the shared code is kept informative by a reconstruction from $\mathbf{z}$ alone and by a prior on $\mathbf{w}$ centred on the class-wise means of $\mathbf{z}$. Our intrinsic path is the analogue of the first. We have no analogue of the second: our prior on $\mathbf{w}$ never references $\mathbf{z}$, and the prior on $\mathbf{z}$ is a standard normal, so nothing else in the objective rewards $\mathbf{z}$ for being informative. The intrinsic path supplies that pressure directly, since under it $\mathbf{z}$ must explain the cell given only the expected response. It runs through the same leakage mixture as the first term, so it never asserts $\kappa = 0$ and contradicts it.

A.3 The Invariance Term as Posterior Regularisation

Neither Pen nor Adv_i is part of any bound. Both are posterior regularisation in the spirit of Ganchev et al. (2010): a constraint on the variational family, $I(\mathbf{z}; [\mathbf{y}, \Phi] | t) = 0$, written as a penalty on the variational objective. This is the same status Slavutsky et al. (2026) give their adversarial term. It is a statement about which q are admissible, not a modification of the generative model, which does not model $\mathbf{y}$ or Φ at all; the constraint states what the decomposition is meant to satisfy, and it acts on the aggregate of the encoder’s outputs over cells rather than on each cell’s posterior, so it is posterior regularisation in a looser sense than the per-instance expectation constraints of Ganchev et al. (2010).

A.4 The Gaussian Conditional Mutual Information

For jointly Gaussian $(\mathbf{z}, \mathbf{v})$ with covariance $\Sigma_{[\mathbf{z}, \mathbf{v}]}$ and marginal covariances $\Sigma_{\mathbf{z}}, \Sigma_{\mathbf{v}}$, the mutual information is $I(\mathbf{z}; \mathbf{v}) = \frac{1}{2}[\log \det \Sigma_{\mathbf{z}} + \log \det \Sigma_{\mathbf{v}} - \log \det \Sigma_{[\mathbf{z}, \mathbf{v}]}]$ (Cover and Thomas, 2006). Equation (19) applies this per type and averages with the type frequencies, which is the Gaussian estimate of $I(\mathbf{z}; \mathbf{v} | t)$ under a type-wise Gaussian assumption. Two properties used in Section 2.5 follow directly. First, the value is invariant to a per-dimension rescaling $\mathbf{u} \mapsto \mathbf{D}\mathbf{u}$ with $\mathbf{D}$ diagonal: each log-determinant gains $\log \det$ of the corresponding block of $\mathbf{D}^2$, and the blocks of $\mathbf{D}$ for $\mathbf{z}$ and $\mathbf{v}$ together make up $\mathbf{D}$, so the three terms cancel exactly. The penalty may therefore be computed on correlation matrices. Second, on the simplex $\sum_k y_k = 1$ the covariance of the full composition vector is singular and the log-determinant is $-\infty$; dropping one column removes the constraint without changing the information, since the dropped column is a deterministic function of the others.

A.5 Symmetries of the Response

Proof of Proposition 1. Write $\mathbf{u}_i = a(\mathbf{z}_i) + \mathbf{B}\mathbf{w}_i$ for the logits of Equation (4). The terms of J that involve a , $\mathbf{B}$, $\mathbf{w}$ or m_ψ are the two reconstruction terms, which are expectations of functions of the logits of cell i and, through $\bar{\rho}_i$, of its neighbours, and the prior divergence of $\mathbf{w}$. The $\mathbf{z}$ -divergence and the invariance penalty depend only on $q(\mathbf{z}_i)$ and t_i and are untouched by all three maps.

(a) $\mathbf{B}\mathbf{Q}^\top \mathbf{Q}\mathbf{w}_i = \mathbf{B}\mathbf{w}_i$, so every draw has the same logits, hence the same ρ_i , $\bar{\rho}_i$ and $\mathbf{p}_i$. Under the prior of Equation (3), $\mathbf{Q}\mathbf{w}_i \sim \mathcal{N}(\mathbf{Q}m_\psi, \sigma_w^2 \mathbf{Q}\mathbf{Q}^\top) = \mathcal{N}(\mathbf{Q}m_\psi, \sigma_w^2 \mathbf{I})$, which is the mapped prior, so the distribution of the counts is unchanged. In J , the mapped posterior and the mapped prior are the images of the originals under $\mathbf{w} \mapsto \mathbf{Q}\mathbf{w}$, so both reconstruction terms, the second of which draws $\tilde{\mathbf{w}}_i$ from the prior, are unchanged, and the divergence between two distributions is invariant under a common bijection. For a signed permutation, $\mathbf{Q} \text{diag } \nu^2 \mathbf{Q}^\top$ is the diagonal matrix with the entries of ν^2 permuted.

(b) The logits become $\mathbf{u}_i + (c(\mathbf{z}_i) + \mathbf{b}^\top \mathbf{w}_i)\mathbf{1}$, a shift by a scalar that is the same for every gene, and the softmax is invariant under such a shift. Every clean composition is therefore unchanged for every draw, and no other term of J involves a or $\mathbf{B}$.

(c) For $\mathbf{z}$ in the support of $q(\mathbf{z}_i)$, $\tau(\mathbf{z}) = t_i$, so the logits become $a(\mathbf{z}) - \mathbf{B}\delta_{t_i} + \mathbf{B}(\mathbf{w} + \delta_{t_i}) = a(\mathbf{z}) + \mathbf{B}\mathbf{w}$, both for posterior draws, which are shifted with the posterior mean, and for prior draws $\tilde{\mathbf{w}}_i$, which are shifted with m_ψ . The same holds for every neighbour, so $\bar{\rho}_i$ is unchanged. The divergence between two Gaussians with the same covariances depends on their means only through their difference, and both means are shifted by δ_{t_i} .

Rescaling. With $\lambda_k = S_{kk}^2$, dimension k of the divergence between $\mathcal{N}(\mu_{w,i}, \text{diag } \nu_{w,i}^2)$ and $\mathcal{N}(m_\psi, \sigma_w^2 \mathbf{I})$ changes under the map by $\frac{1}{2}[(\lambda_k - 1)A_{ik} - \log \lambda_k]$ with $A_{ik} = (\nu_{w,ik}^2 + (\mu_{w,ik} - \psi_{ik})^2)/\sigma_w^2$, where ψ_{ik} is coordinate k of $m_\psi(\mathbf{c}_i, t_i)$. Because σ_w is fixed, for $\lambda_k \neq 1$ this vanishes only where $A_{ik} = \log \lambda_k / (\lambda_k - 1)$, not in general, so the map does not preserve J .

Readouts. Under (a) the differences $\mathbf{w}_i - \bar{\mathbf{w}}_t$ and $m_\psi(\mathbf{c}_i, t_i) - m_\psi(\bar{\mathbf{c}}_t, t_i)$ are multiplied by $\mathbf{Q}$ and $\mathbf{B}$ by $\mathbf{Q}^\top$ on

the right, so their products are unchanged. Under (b) each product gains $\mathbf{1b}^\top(\cdot)$, a constant over genes, which centring removes. Under (c) both terms of each difference are shifted by the same δ_t , and the contexts are unchanged. $\square$

A.6 Feedback Amplification Through the Leakage Path (Heuristic)

The following is a first-order heuristic, not a theorem. Write the decoder output of cell j as ρ_j and suppose gradients were allowed to flow from cell i 's reconstruction through $\bar{\rho}_i = \sum_j \beta_{ij} \rho_j$ into the parameters that produce ρ_j . Linearising around a fixed point of the coupled system, a perturbation δ of a cell's clean composition induces a perturbation $\kappa \beta \delta$ of its neighbours' fitted compositions, which feeds back through the same path; the total response is $\delta \sum_{n \geq 0} (\kappa \mathbf{P}_\beta)^n$ for a row-stochastic $\mathbf{P}_\beta$ built from β , whose spectral radius is at most one, so the geometric series is bounded by $1/(1 - \kappa)$. The stop-gradient truncates the series at $n = 0$. The argument concerns perturbations of the fitted compositions; the gradient fixed point through shared decoder parameters is more involved, and we have not fitted the open-path model; Ergen and Yosef (2025) close the same path for training stability.

B DESIGN RATIONALE

This appendix carries the parts of the arguments of Section 2 that the main text only summarises. It does not repeat what is already there.

B.1 The Mirror Attractor

The attention layer of Equation (10) computes logits $e_{ij} = \mathbf{a}^\top \text{LeakyReLU}(\mathbf{W}_{\text{dst}} \mathbf{h}_i + \mathbf{W}_{\text{src}} \mathbf{h}_j)$ and aggregates $\sum_j \pi_{ij} \mathbf{W}_{\text{src}} \mathbf{h}_j$, so the destination feature $\mathbf{h}_i$ enters the *attention* but never the *value* (Brody et al., 2022). With $\mathbf{z}_i$ as the destination feature, the selection channel of Section 2.3 opens. What the type query gives up is state-specific attention, which belongs in the decoder (as type-dependent loadings or a low-rank interaction with $\mathbf{z}$) rather than in the context encoder.

The diagnostic is the within-type coefficient of determination of $\mathbf{z}_i$ regressed on $\mathbf{c}_i$ against a within-type permuted control (Section 2.7). The correlation between π_{ij} and $\text{sim}(\mathbf{z}_i, \mathbf{z}_j)$ would serve under a $\mathbf{z}$ query; with the type as query and types as sources the attention weights depend on the types alone (Equation (27)), and the correlation is vacuous. Downstream, a mirror is indistinguishable from ordinary posterior collapse of $\mathbf{w}$; a free-bits remedy (Kingma et al., 2016) would mask it and leave the anomaly score meaningless, which is why the mirror is ruled out first.

Zeroing $\mathbf{z}_i$ while keeping self-loops is not an alternative: the softmax still spends attention mass on the self-edge, down-weighting neighbours, and under $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ the origin is the prior mean, so a zero vector asserts “this cell is average” rather than “ignore this cell”.

B.2 Why the Context Gets No Per-Edge Geometry, and Why the Graph Is Delaunay

A discretised contact indicator (touching, near, far) looks as if it dodges the collinearity objection of Section 2.3, but on this platform, whose segmented polygons rarely touch, whether two of them touch is largely a measurement of local density: dense regions touch, sparse regions do not. The indicator would smuggle density back in, in the confounded direction, and a tolerance parameter only moves the threshold. Figure 2 shows it on one field: exact contact leaves most cells without a neighbour, a $1 \mu\text{m}$ tolerance connects the dense tissue but leaves the separated cells of the sparse tissue isolated, and Delaunay adjacency connects them all.

Delaunay adjacency resolves it: two cells are neighbours if and only if their Voronoi cells share a face, which is the correct notion of adjacency for a packing, built from centroids alone. The one parameter that returns is the pruning of long edges, and it is much less sensitive than choosing a neighbour count. The Voronoi face length is used in the kernel rather than shared segmented boundary because the Voronoi face is where the boundary between two cells effectively lies, its length is a contact-extent proxy defined whether or not the polygons touch, and it is independent of the expansion setting (Figure 3).

Table 3 and Figure 4 make the comparison concrete on every section. The sections were segmented mostly by an interior RNA stain rather than by nuclear expansion, and adjacent polygons still rarely touch. A contact

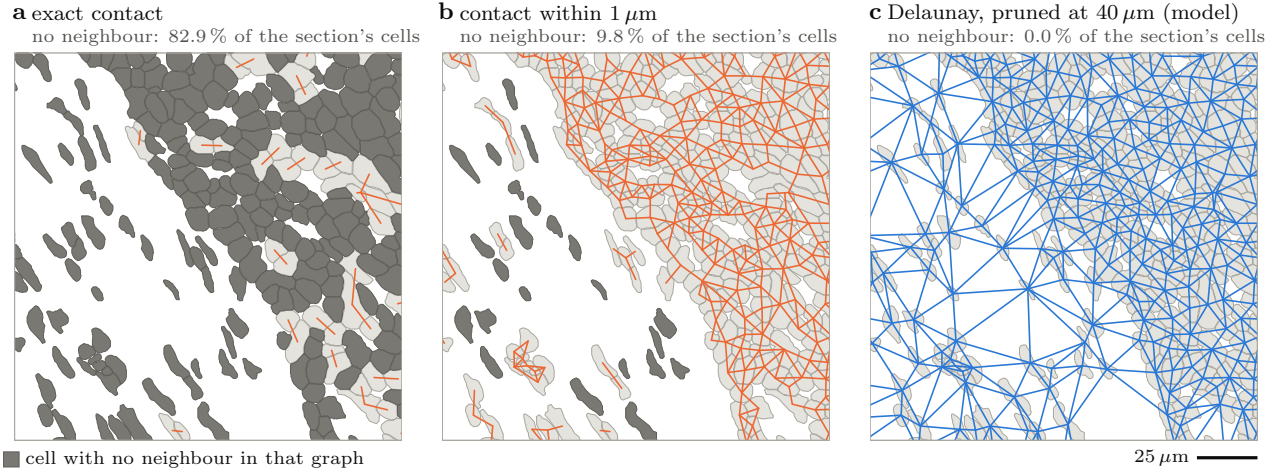


Figure 2: Three neighbour graphs on one 160 μm field of the primary section, each a subset of the model’s edge set. **a**, Edges whose segmented polygons touch. **b**, Edges along which the two polygons run within 1 μm of each other, the contact of Table 3. **c**, The model’s graph: Delaunay adjacency pruned at 40 μm . Cells with no edge in a panel’s graph are dark. The share above each panel is over the whole section, among cells with at least one edge in c. The field is chosen to be representative: among fields that contain both sparse and dense tissue, it is the one whose shares under a and b are closest to the section’s. Edges are drawn at one width; their weights in β are illustrated in Figure 1a. All panels share one scale.

kernel is therefore zero on about half of the edges, and it leaves a density-dependent share of cells with no leakage at all: up to a third of cells in the sparsest fifth of a section, almost none in the densest. Whether a cell can receive leakage would then be decided by local density, which is part of the niche. The Voronoi kernel gives every connected cell the same leak budget κ . Both kernels give nearer neighbours more weight; for β that dependence is the one the distance decay introduces by design, and the face does not add to it.

One consequence of the softmax over a Delaunay neighbourhood should be stated. With the type as query and one-hot types as sources, the logit of edge ij depends on (t_i, t_j) alone, say $s_{t_i t_j}$, and grouping the neighbours by type gives, per head,

$$\mathbf{g}_i = \frac{\sum_k y_{ik} e^{s_{t_i k}} \mathbf{W}_{\text{src}} \mathbf{e}_k}{\sum_k y_{ik} e^{s_{t_i k}}}, \quad (27)$$

with $\mathbf{e}_k$ the k -th unit vector: a learned, type-specific reweighting of the neighbour composition in which the degree cancels. The context therefore depends on the fractions $\mathbf{y}_i$, not on the counts. It distinguishes one tumour neighbour among six from five among six, but not a neighbourhood from the same neighbourhood with every type count scaled up together; in particular a single-type neighbourhood gives the same context whatever its size. On this tessellation the degree is nearly constant across the section, so a fraction determines a count and the blindness costs little. An attention sink, a fixed null logit that joins every softmax and restores a saturating dependence on the count, is implemented and was tested over three seeds. It learned no dose response, placing its half-point below a single neighbour, did not improve the held-out transport of the response, and in two of the seeds made the prior mean read the context less faithfully. It is therefore left off. It would become meaningful only on a graph whose degree varies, where the invariance target would have to grow with it.

The fallback is empirical. If m_ψ predicts responses poorly, or held-out reconstruction is worse than a comparison run whose attention layer carries edge features (distance, face length), that is direct evidence that the continuous signal mattered. If it is added back, the correlation between π_{ij} and β_{ij} must be monitored: a high correlation means the attention has rediscovered the leakage kernel and $\mathbf{w}$ is becoming collinear with $\bar{\rho}$. That risk is tolerable while κ is fixed, since there is no free contamination parameter to feed, and intolerable once κ is estimated. **[PENDING RUNS: the edge-feature comparison run has not been performed]**

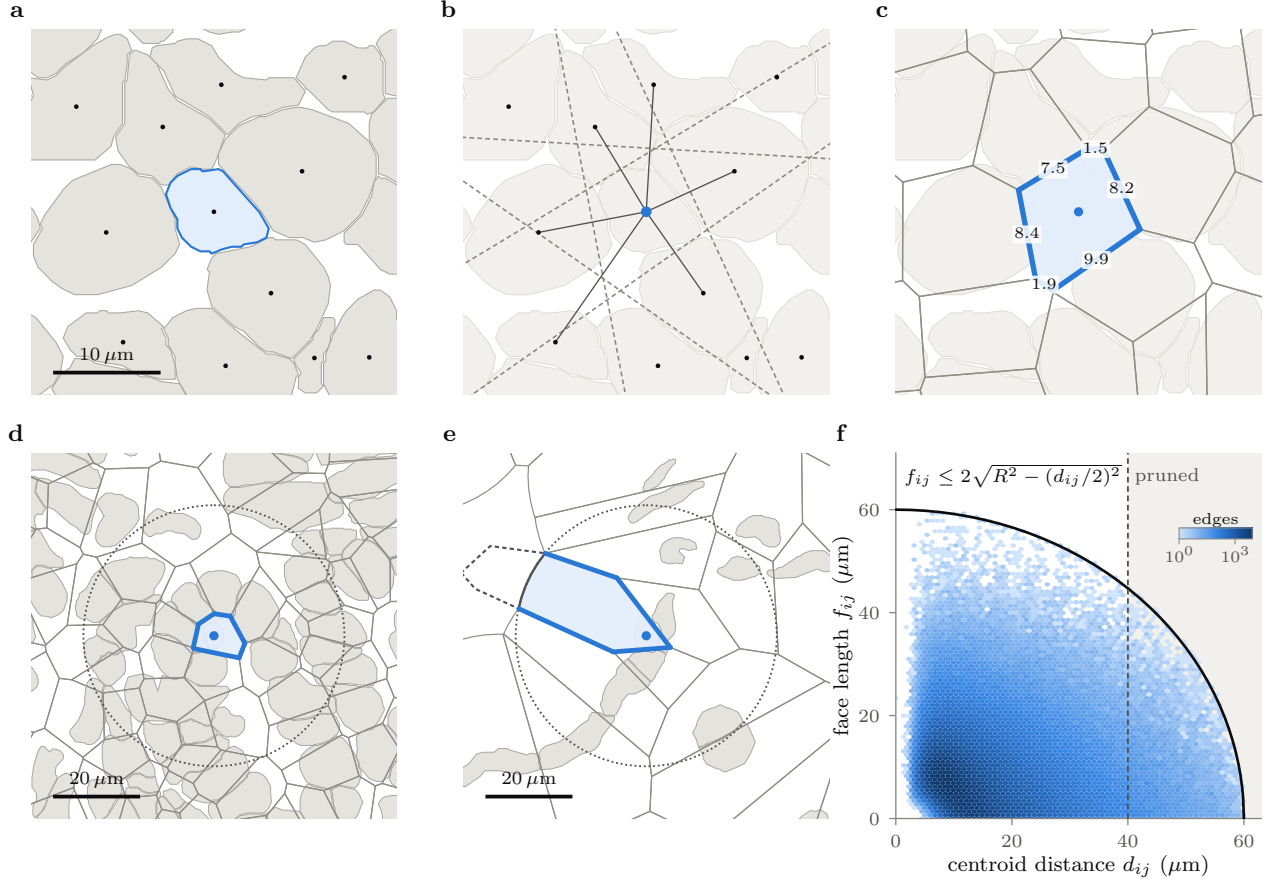


Figure 3: How the Voronoi face is built, and what bounds it, on the primary section. **a**, Segmented polygons and their centroids around a typical cell (blue). **b**, The polygons are set aside; each neighbour contributes the perpendicular bisector (dashed) of the segment joining the two centroids. **c**, The bisectors cut one another, and what survives between the cell and a neighbour is the face f_{ij} (blue, length in μm); the shaded area is the cell’s Voronoi region. Panels a–c share one scale. **d**, A dense neighbourhood and **e**, a cell at a tissue border, at one scale, each with the $30\mu\text{m}$ clip disc (dotted). At the border the unclipped region (dashed) runs into the gap, the disc bounds it, and the dark stretch of the outline between two faces is the clip arc. **f**, Face length against centroid distance for every edge of the section. Clipping bounds each face by the chord $2\sqrt{R^2 - (d_{ij}/2)^2}$ of the disc, with $R = 30\mu\text{m}$, and removes every edge with $d_{ij} \geq 2R$; the model keeps edges up to $40\mu\text{m}$ (unshaded).

B.3 The Image Descriptor and Its Mask

The image descriptor Φ_i is the embedding of a $128\mu\text{m}$ field around the cell (256 pixels at $0.5\mu\text{m}$ per pixel) in the section’s four morphology channels (Figure 1b), by KRONOS, a frozen foundation model for multiplexed tissue images (Shaban et al., 2025): its first release, a ViT-S/16 with a 384-dimensional embedding. The channels map onto the model’s marker vocabulary only approximately: the membrane and αSMA channels are antibody mixes read as their named component, and 18S RNA lies outside the vocabulary and is given an unused marker identity. The cell is removed by setting a disc around it to zero, not its own polygon: a polygon-shaped hole would keep the cell’s outline, the most type-informative feature of all. The disc is centred on the crop anchor and has one radius for every cell, so the hole carries no information about the cell it removes, and it is filled with zeros, a constant artefact that the image model encodes identically for every cell. Its radius is the smallest that covers essentially every cell (Figure 5). A cell’s covering radius, the radius of the smallest disc centred on its anchor that contains its polygon, has a long tail of elongated smooth-muscle cells; a $25\mu\text{m}$ disc leaves six cells of the primary section outside it and masks 12% of the field. Those six cells were left out of the comparison below, and in the model their image descriptor is set to zero. A cell that sticks out of the disc would leak the identity the disc exists to remove, so partial coverage is not an option. At this density the disc also removes the cell’s first ring or

Table 3: Segmented-polygon contact as a leakage kernel, against the Voronoi-face kernel β , on every section and on the model’s edge set (Delaunay adjacency pruned at $40\mu\text{m}$). Both kernels use the same distance decay $\exp(-d_{ij}/\tau)$ and the same row normalisation. The contact kernel weights an edge by the length of membrane along which the two segmented polygons run within $1\mu\text{m}$ of each other; β weights it by the Voronoi face. FF: fresh frozen. *Touching*: adjacent pairs whose polygons are at zero distance. *No-leak cells*: connected cells whose kernel row is all zero, so that they receive no leakage at any κ ; *sparse* and *dense* are the lowest and highest fifth of cells by the number of centroids within $20\mu\text{m}$. ρ : Spearman correlation, over edges, between a neighbour’s share of the row and the gap between the two polygons. The Voronoi kernel has no zero-weight edges and no no-leak cells on any section. Its dependence on the gap is that of the distance decay alone (shares from the decay alone: ρ between -0.49 and -0.62).

Section	Segmented by (% cells)			Edges (%)		No-leak cells, contact (%)			ρ (share, gap)	
	interior	boundary	nucleus	touching	no contact	all	sparse	dense	contact	β
Ovarian cancer (FFPE)	69	29	2.0	3.5	47	9.8	33.6	0.6	-0.82	-0.48
Lung cancer (FFPE)	82	16	2.2	2.7	57	11.1	27.4	0.7	-0.84	-0.55
Ovarian cancer (FF)	70	28	2.5	6.7	38	3.2	10.6	0.1	-0.80	-0.52
Lung TMA, section A	87	12	1.0	2.2	47	6.3	24.8	0.1	-0.84	-0.54
Lung TMA, section B	87	12	1.2	2.2	47	6.2	24.6	0.1	-0.84	-0.53

Table 4: What each mask leaves in the image embedding, on the primary section: macro one-versus-rest AUC of a multinomial logistic regression predicting the cell’s lineage label (the labels the model is fitted with; 11 classes, Unassigned cells excluded as targets but counted as neighbours) from the embedding, on a spatial split (1 mm tiles, cells within half a field of a tile edge dropped; 40,000 training and about 74,000 test cells), for the two releases of the image model, KRONOS and KRONOS2 (Shaban et al., 2025). The first row predicts the label from the neighbours’ labels alone, with no image. The model uses KRONOS with the disc zeroed (bold).

Input to the classifier	KRONOS	KRONOS2
Neighbours’ labels, no image	0.918	
Whole patch	0.881	0.878
Disc over the cell zeroed	0.857	0.846
The cell alone	0.905	0.919
The cell alone, native resolution	0.923	–

two, so Φ_i describes the tissue beyond roughly $25\mu\text{m}$.

Table 4 shows what each mask leaves in the embedding, read by a multinomial logistic regression that predicts the cell’s lineage label from it. The split is spatial, in tiles with a margin, because the fields of neighbouring cells overlap almost completely and a random split would put the same pixels on both sides. Three readings follow. The cell alone carries more type information than the whole patch: the surrounding tissue dilutes it rather than adding to it. Removing the cell costs little, because the patch embedding is dominated by its surroundings (Figure 6a, b). And the masked patch predicts the label less well than the neighbours’ labels alone, so what it knows about the cell is what type-homogeneous neighbourhoods imply, which is what a descriptor of the niche should carry. On the masked patch the first release does at least as well as its successor KRONOS2 (a ViT-B/16 with a 768-dimensional embedding, from the same authors); KRONOS2 is better only on the cell alone, where the model deliberately does not look. We therefore use the first release: its embedding is half as wide (384 against 768 dimensions), which keeps the image part of the context descriptor small, and it embeds cells about five times faster.

B.4 Three Routes to a Vanishing Response Divergence

Cause	Check	Fix
Posterior collapse	α_w large relative to reconstruction	lower α_w ; free bits
Mirror	within-type R^2 of $\mathbf{z}_i$ on $\mathbf{c}_i$ vs. permuted control	type embedding as query; types only as sources
Starved of ego evidence	does the response posterior receive $\mathbf{x}_i$ directly?	add the direct path

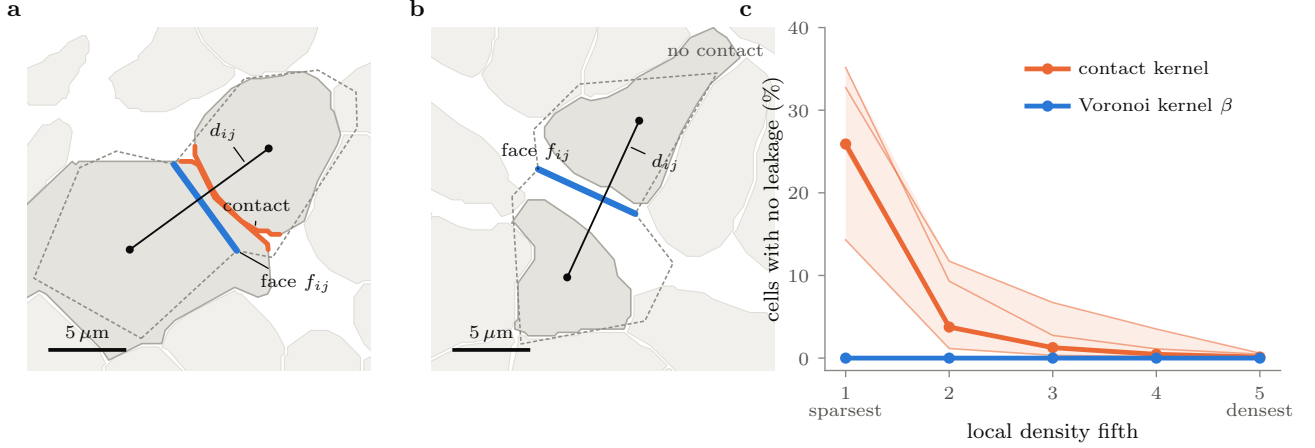


Figure 4: Why the leakage kernel is built on Voronoi faces rather than on segmented contact. **a**, An edge whose segmented polygons touch and **b**, an edge whose polygons do not, each the edge of its kind closest to the primary section’s median centroid distance and median face length. The two have almost the same centroid distance d_{ij} and face f_{ij} (blue), which is all the kernel β uses; contact (orange), the stretch of each cell’s membrane within $1\mu\text{m}$ of the other cell, is present in a and absent in b. Dashed lines are the two cells’ Voronoi regions; both panels share one scale. **c**, The share of cells that a contact-weighted kernel would leave with no leakage at all, by fifth of local density (centroids within $20\mu\text{m}$), on each section of Table 3: the median (bold), the range (band) and each section (thin). Both kernels use the model’s edge set, the same distance decay and the same row normalisation; the Voronoi kernel leaves no connected cell without leakage on any section.

B.5 Conditional Invariance and Label Granularity

The marginal-invariance failure is documented by Slavutsky et al. (2026) for the conditional subspace VAE (Klys et al., 2018), whose invariance drives the agreement between the shared code and the nuisance variable to zero while collapsing its agreement with cell type [TODO: quote the exact numbers from that comparison]. Lineage-level labels are preferable to finer or state-defined ones for four reasons: fine labels leave $\mathbf{z}$ with nothing to carry, they block more of the position $\rightarrow$ fate $\rightarrow$ expression path, they destroy the overlap that any conditional comparison needs, and, because t_i is itself a function of $\mathbf{x}_i$, hence of leakage and of any response that moves a cell across a label boundary, conditioning on it conditions on a descendant of the niche, so that even the true intrinsic state can depend on $\mathbf{y}_i$ given t_i near label boundaries (Hernán et al., 2004).

B.6 The Leak Fraction: Grid Range and Upgrade Path

The grid range comes from three sources: zero as anchor; published misassignment estimates for this platform [NEEDS A SOURCE]; and a marker-set ceiling. For an atlas-defined set $\mathcal{M}$ of genes that a cell of type t_i should not express, and an empirical neighbour profile $\tilde{\rho}_i$ obtained by averaging the observed compositions of i ’s neighbours with the kernel β , the ratio $\sum_{g \in \mathcal{M}} x_{ig} / (\ell_i \sum_{g \in \mathcal{M}} \tilde{\rho}_{ig})$ bounds the leak fraction from above. The set must be defined externally: ranking genes by their neighbour-to-own expression ratio from the same counts selects the zero-count genes and drives the bound to zero with probability one. [PENDING RUNS: compute the marker-set ceiling on both sections]

This version of the model sweeps κ ; it cannot estimate it. Assigning transcripts to cells by position does not settle it either: counts of transcripts that lie nearer a neighbour’s nucleus than the cell’s own are dominated by segmentation geometry, since a large cell beside small ones counts its own far-side transcripts as foreign, so such counts bound leakage from above but do not identify how it scales with the depth of the receiving or the donating cell. Estimation needs two views of one cell that share a foreign profile at different contamination levels, which a nuclear/extranuclear split of the transcript coordinates provides. Its identifying condition is that gene-wise nuclear retention varies across genes, which is also a reason to expect the gene-specific leakage that the main model does not represent (Section 2.2), and which a two-view likelihood would be the natural place to model; with constant retention both views lie on the same line between ρ_i and $\tilde{\rho}_i$ and nothing is pinned. Adding it does not restructure the model: the likelihood becomes a multinomial over $2G$ bins and κ moves from hyperparameter

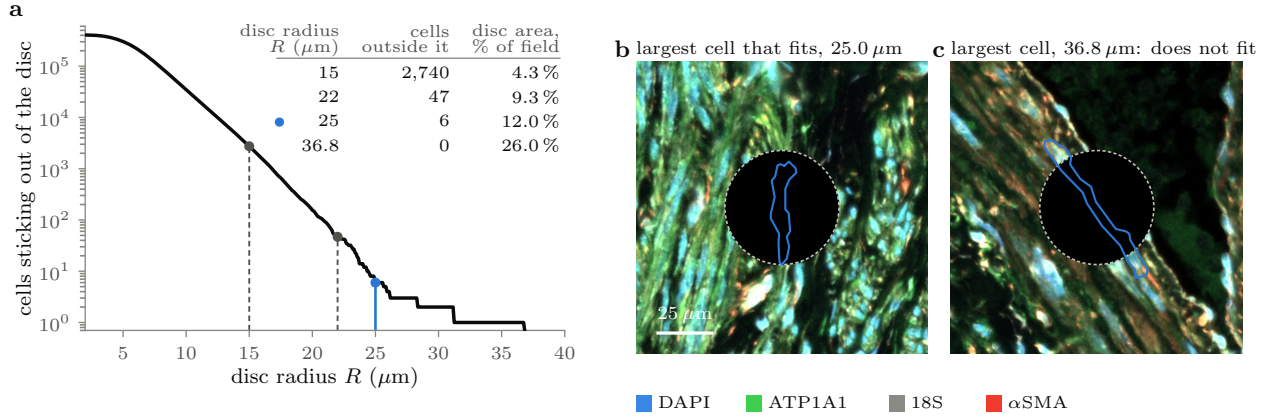


Figure 5: Why the mask is a 25 μm disc, on the primary section. **a**, The number of cells that would stick out of a disc of radius R centred on the crop anchor, from each cell’s covering radius; the table gives the radii considered, the cells left outside, and the share of the 128 μm field each disc masks. The last row is the largest cell on the section. **b**, The largest cell that fits the chosen disc and **c**, the largest cell on the section, which does not, both drawn as the image model receives them: four channels, disc set to zero, with the cell’s outline for reference. Both are elongated smooth-muscle cells.

to latent. We leave it as future work.

What the data say about κ can be stated exactly at the level of compositions.

Proposition 2. *Hold the neighbours’ clean compositions, and hence $\bar{\rho}_i$, fixed, and let $\mathbf{p}_i = (1 - \kappa)\boldsymbol{\rho}_i + \kappa\bar{\rho}_i$ with $\boldsymbol{\rho}_i$ in the simplex. A leak fraction $\kappa' \in [0, 1]$ reproduces $\mathbf{p}_i$ with a clean composition in the simplex if and only if $\kappa' \leq \bar{\kappa}_i = \min_{g: \bar{\rho}_{ig} > 0} p_{ig}/\bar{\rho}_{ig}$. The true κ satisfies $\kappa \leq \bar{\kappa}_i$, with equality if and only if $\rho_{ig} = 0$ for some gene g with $\bar{\rho}_{ig} > 0$.*

Proof. The clean composition implied by κ' is $\boldsymbol{\rho}_i(\kappa') = (\mathbf{p}_i - \kappa'\bar{\rho}_i)/(1 - \kappa')$, which sums to one for every κ' and is non-negative in gene g if and only if $\kappa'\bar{\rho}_{ig} \leq p_{ig}$; this holds for every gene if and only if $\kappa' \leq \bar{\kappa}_i$. Since $p_{ig}/\bar{\rho}_{ig} = \kappa + (1 - \kappa)\rho_{ig}/\bar{\rho}_{ig} \geq \kappa$, the true value lies below the bound, and meets it exactly when a gene the neighbours express has zero share in the cell’s own composition. $\square$

The data therefore bound κ from above and never from below, and a section-wide κ must respect the bound of every cell. They pin it only through a gene the cell cannot express itself, which is what the marker-set ceiling above and a nuclear/extranuclear split would supply. Figure 7 draws both cases. The statement is about compositions: with counts the bound is soft, since a κ above it charges each cell for neighbour transcripts it does not show, and the decoder’s form can narrow the interval further, which is the route that expected low-dimensionality takes and that a response to the niche weakens (Section 1).

Definition 1 is easiest to read on an example. A readout is any number reported from a fit, for instance how much a fibroblast programme differs between fibroblasts next to tumour and fibroblasts in stroma. At $\kappa = 0$ the model assigns all of that difference to the cells themselves. Refitting at each larger κ attributes more of the resemblance between neighbours to leakage, so a difference that is only leakage shrinks toward its null value, and one that reflects biology persists. The breakdown point κ^* is the smallest leak fraction at which the readout can no longer be told from its null, or changes sign: how much leakage it would take to explain the readout away. Figure 8 shows the four cases. A readout that survives every grid value cannot be explained by leakage of the modelled form up to the top of the grid (a). A magnitude may shrink as κ grows without breaking down (b): it shrinks by construction, because the mixture takes over part of the resemblance between neighbours, and it stands as long as it stays above its null. A readout that breaks down inside the grid is explained away by that much leakage (c), and one that breaks down at the first grid point above zero is explained away by any leakage at all (d). The higher the breakdown point, the stronger the finding. Read with Figure 7, the two say what a finding needs: the data bound the leak fraction from above, and a readout whose breakdown point lies above that

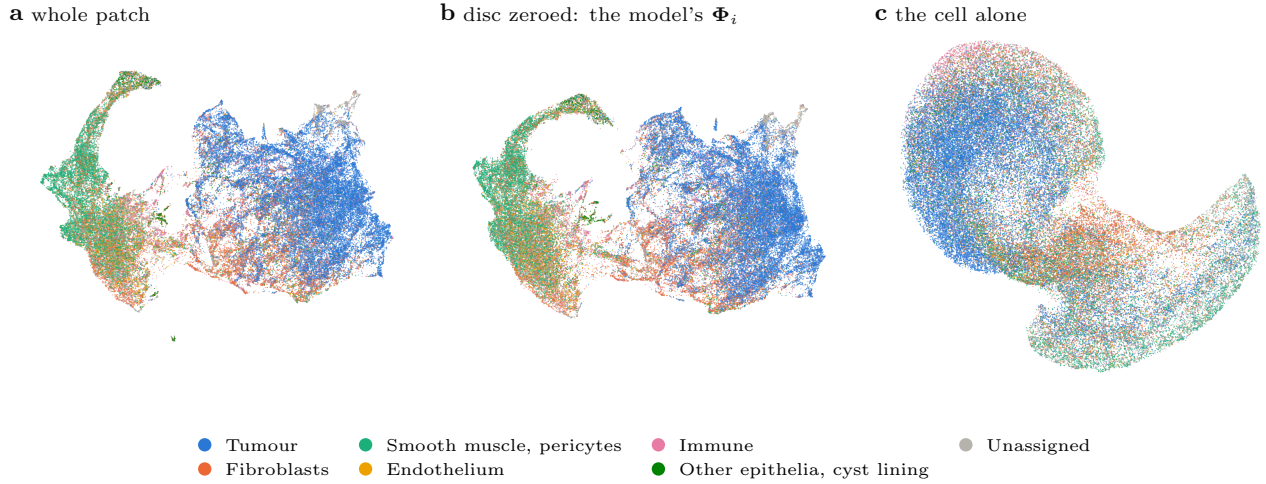


Figure 6: The KRONOS embedding (first release, the one the model uses) under three masks, as UMAPs of the same 60,000 cells of the primary section (principal components, then UMAP). **a**, The whole $128\mu\text{m}$ patch. **b**, The disc over the cell zeroed, the descriptor Φ_i the model uses; it differs little from **a**, because the patch is dominated by the surrounding tissue. **c**, Everything outside the cell’s polygon zeroed, what the cell alone carries. Colour is the section’s lineage label, the one the model is fitted with, with the smaller lineages grouped for display. Unassigned includes a low-depth class of mixed lineage that the section’s original annotation had called a tumour subtype.

bound survives every leak fraction the data allow.

B.7 Learned Variances: What to Learn and What Not To

If a conditional $\sigma_w(\mathbf{c}, t)$ is wanted, because it would standardise deviation against local variability so that a deviation of two in a niche where everyone deviates by two does not read as anomalous, it must be bounded, e.g. $\sigma_w \in [0.5, 2]$ through a scaled sigmoid, so that it cannot inflate away, and its mean over cells must be checked for upward drift over training.

The prior on $\mathbf{z}$ is a standard normal. A richer prior, a mixture with one component per type or a VampPrior with archetypal pseudo-inputs (Tomczak and Welling, 2018), would be a genuine upgrade for cell-type structure and would make t a latent rather than an input, but that is a different change, not a variance question.

C IMPLEMENTATION DETAILS

C.1 Architecture and Hyperparameters

Table 5 lists every network and every fixed quantity as implemented; the weights marked calibrated are set per section by the protocol of Section 2.5 and are not quoted here. All multilayer perceptrons use rectified-linear hidden units.

C.2 Choices Made in Implementation

Where the model description of Section 2 leaves a choice open, Table 6 records what was done and why. Every entry is also stated in the method text; the table collects them in one place. **[TODO: author to decide whether this table stays; it duplicates the method text]**

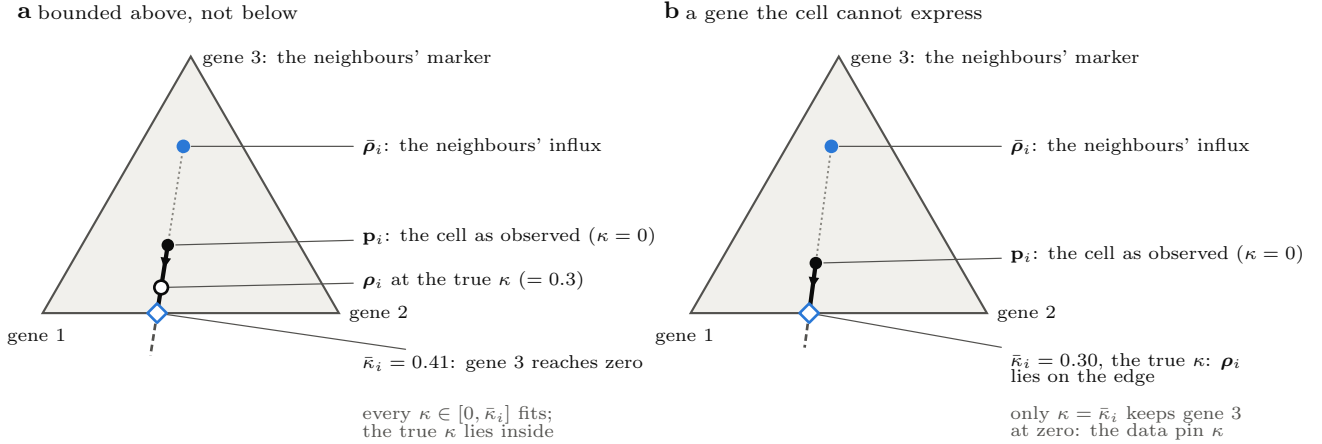


Figure 7: What the data say about κ , on the simplex of three genes; schematic, with compositions chosen for legibility. With the neighbours' influx $\bar{\rho}_i$ held fixed, each κ implies the clean composition $\rho_i(\kappa) = (\mathbf{p}_i - \kappa \bar{\rho}_i) / (1 - \kappa)$: a line that starts at the observed composition $\mathbf{p}_i$ ($\kappa = 0$) and runs away from $\bar{\rho}_i$ (arrow). It is valid until it leaves the simplex at $\bar{\kappa}_i$ (diamond) and invalid beyond (dashed). **a**, A cell that expresses a little of the neighbours' marker itself: every $\kappa \in [0, \bar{\kappa}_i]$ reproduces $\mathbf{p}_i$ exactly, and the true value lies inside. **b**, A cell that cannot express the marker: its true composition lies on the edge, the line leaves the simplex exactly there, and $\bar{\kappa}_i$ is the true κ .

C.3 Tiles and the Two-Hop Halo

Figure 9 shows the tiles, the held-out split and the halo of one training tile on the primary section, as built for a fit (Section 2.6).

C.4 The Held-Out Probe

Figure 10 shows how the probe of Section 2.5 is fitted, scored and read.

Table 8 gives the probe on the final fits of every section.

D EXPERIMENTAL DETAILS

D.1 Sections

Table 9 lists the sections of Section 3.1, and Figure 11 shows them. The lineage classes include Unassigned. On the ovarian FFPE section it also holds a low-depth class of mixed lineage that the curated annotation had named as a tumour subtype, which is why its share is the largest.

D.2 Lineage Labels

The fitted labels are lineage-level on every section (Section 2.1). States, such as proliferating, activated or hypoxic, and locations, such as tumour- or stroma-associated, are merged into their lineage; lineages and developmentally distinct sub-lineages are kept. The mappings were fixed and recorded before any model was refitted.

Curated annotations. On the ovarian FFPE section the eighteen curated classes become twelve (Table 10). Two classes needed a decision. The class named as a SOX2-OT⁺ tumour subtype is mostly not tumour: its cells have a median of 27 transcripts against 333 for tumour cells, three quarters show no SOX2-OT count, and its pooled profile decomposes mostly into macrophage expression, so it is folded into Unassigned rather than into the tumour lineage. The class named as malignant cells lining a cyst expresses mesothelial markers (CALB2, PRG4, BNC1, PDPN, UPK3B) and lacks the tumour markers (EPCAM, PAX8, ESR1), and becomes its own lineage. On the lung TMA the thirty-five curated classes become thirty: the state classes (activated, inflammatory and

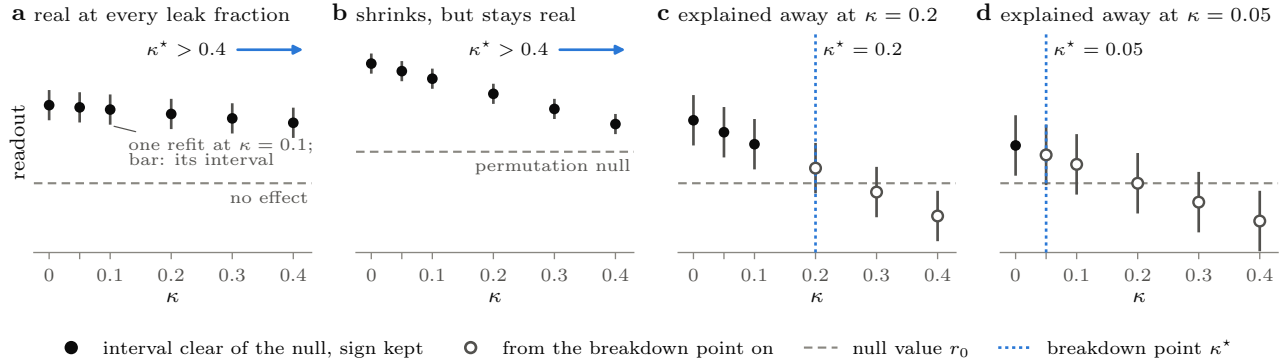


Figure 8: The breakdown point of Definition 1, on four schematic readouts: four different candidate findings, not four ways of choosing κ , which is never chosen. The curves are illustrative, not fitted, and only the grid of κ is the model’s. Each panel shows a readout at every grid value with its interval (bars) and its null value r_0 (dashed). Filled markers: the interval is clear of the null and the readout keeps the sign it has at $\kappa = 0$; hollow markers: from the breakdown point κ^* (dotted) on. Each point is a separate fit at that κ . **a**, A signed contrast that survives the grid, reported as $\kappa^* > 0.4$. **b**, A magnitude that shrinks as κ grows, as expected by construction because more of the resemblance between neighbours is assigned to the mixture, but stays above its permutation null: it also survives. **c**, A contrast that 20% leakage explains away, $\kappa^* = 0.2$; its sign flips beyond it. **d**, A contrast that breaks down at the first grid point above zero, $\kappa^* = 0.05$: any leakage explains it away, so it is not separable from leakage of the modelled form.

myofibroblasts, activated endothelium and proliferating cells) are reassigned cell by cell to the sub-lineage their markers indicate, with the cell-cycle genes left out for the proliferating class, and the other classes are kept.

Other judgement calls. Four further choices are not forced by the rule and are stated so that they can be revisited. The tumour- and stroma-associated fibroblasts of the ovarian FFPE section are merged, because what separates them is the cancer-associated fibroblast programme, a state, rather than a lineage. Its two endothelial classes are merged too, although what separates them is less clear: a tip-cell state on one side and a weakly venous identity on the other. On the lung TMA the alveolar and adventitial fibroblasts are kept as distinct sub-lineages, and the myofibroblasts, which could belong to either, are split between them by their markers, about evenly. The cell-by-cell reassignment of the TMA’s state classes is expected to be right for about three cells in four, and its margins are recorded with the mapping.

Clustered sections. On the lung FFPE section and the fresh-frozen ovarian section the labels start from unsupervised clusters, and each cluster is assigned the lineage its marker genes indicate, on the lung TMA’s vocabulary for the lung section (Table 11). Clusters of immune cells next to tumour that carry some tumour signal keep their immune lineage, since that signal is the leakage the model is built to absorb, and one low-depth cluster of the fresh-frozen section becomes Unassigned.

D.3 Baselines

Table 12 gives the configuration of each comparison method. Every method is run at its published defaults, and on our pruned Delaunay graph where it accepts an external graph; resolVI and MintFlow build their own. The methods that read a label read the lineage labels. SIMVI does not read one, so refitting it at lineage labels only checks reproducibility.

D.4 Assets and Licences

The ovarian FFPE, lung FFPE and fresh-frozen ovarian sections are public datasets of 10x Genomics (10x Genomics, 2024a,c,b), released under the Creative Commons Attribution 4.0 International licence (CC BY 4.0). The TMA core and its serial section are two samples of GEO series GSE315411 (Williams-Katek et al., 2026); NCBI places no restriction on the use or distribution of GEO data, while noting that submitters may claim

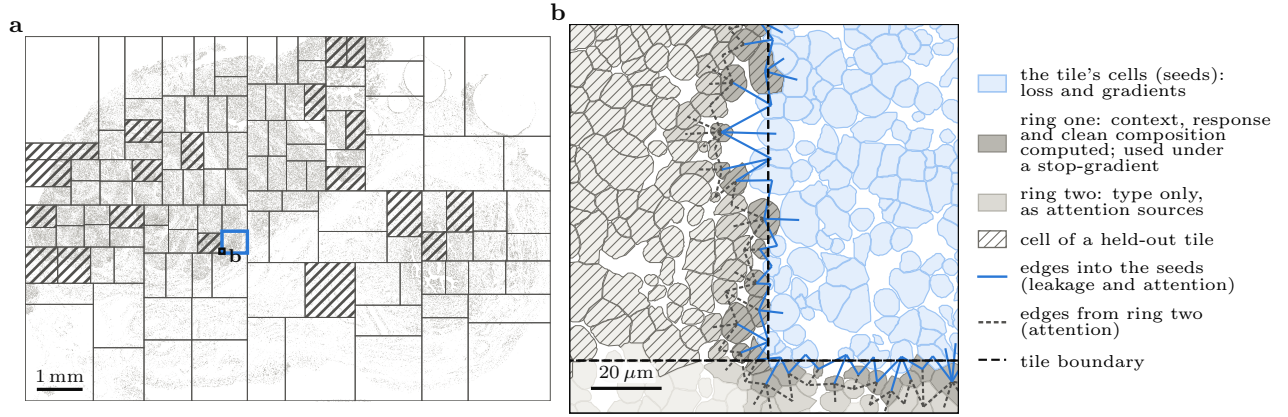


Figure 9: Tiles, the held-out split and the two-hop halo on the primary section. **a**, The section cut into contiguous tiles of at most 4,096 cells by recursive median split. Hatched tiles are held out (15% of tiles, one split). The blue tile is shown in **b**, and the black square marks the window of **b**. **b**, Where that training tile meets a held-out tile (left) and another training tile (below). The tile’s cells are the seeds, on which the loss is evaluated. Ring one, their graph neighbours, has its context, response and clean composition computed in the same pass. Those compositions enter the seeds’ foreign influx under a stop-gradient, and ring one’s types are attention sources for the seeds (solid edges). Ring two, the neighbours of ring one, contributes only its types, as attention sources for ring one (dashed edges). Both rings are read from the leakage kernel’s own edge list. Hatched cells belong to the held-out tile: ring cells drawn from it enter training as inputs, with no loss and no gradient (Section 2.6).

rights in the data they submit [\[TODO: verify the wording on the GEO disclaimer page\]](#). The image model KRONOS and its successor KRONOS2 (Shaban et al., 2025) are released under CC BY-NC-ND 4.0, for non-commercial academic research and after registration; we use them for that purpose and redistribute neither. The comparison methods are released under the BSD 3-Clause licence: scvi-tools, which contains resolVI, SIMVI, MintFlow and Cellina. The code will be released on acceptance; no new data are released.

D.5 Training Cost

Table 13 gives the cost of a fit. DISCELL’s time per epoch is pure training, measured over twenty epochs with no evaluation. Its wall-clock time covers the three final fits, including evaluations every five epochs and the early stop. The comparison methods’ times are whole fits, as recorded when they were run.

In symbols, a training step on a tile of n seeds with n_1 cells in ring one costs $O((n + n_1)Gh)$ in the encoders and decoder, for hidden width h , and $O(|E_n|G)$ in the foreign influx, for the $|E_n|$ edges into the seeds (Section 2.6). The six adversary head updates add $O(6(n + n_1)h_a(d_z + K))$ with $h_a = 64$, small beside the gene-sized terms because d_z and K are far below G . An epoch visits every training tile once, so its cost is linear in the number of cells; the held-out probes are fitted on at most 30,000 cells whatever the section’s size. Peak memory is the section’s count matrix, held resident, plus the gathered edge-by-gene rows of the influx, $O(|E_n|G)$ per step, which the tile size bounds. The leakage sweep multiplies the cost of a fit by its six grid points and three seeds, eighteen fits per section. Sample size is not analysed: how many cells a section needs for the readouts to stabilise is not studied beyond the range of sizes in Table 9.

D.6 Planted Worlds

Three questions about the fitting procedure are answered on simulated sections, where the truth is known.

The amortisation gap: what is measured. The intrinsic encoder $q(\mathbf{z}_i | \mathbf{x}_i, t_i)$ reads a cell’s counts and type but not the foreign influx $\bar{\rho}_i$, although the counts contain it. Its posterior mean can therefore differ from the exact posterior mean $\mathbb{E}[\mathbf{z}_i | \mathbf{x}_i, t_i, \bar{\rho}_i]$ that a model told the influx would reach. This difference, the amortisation

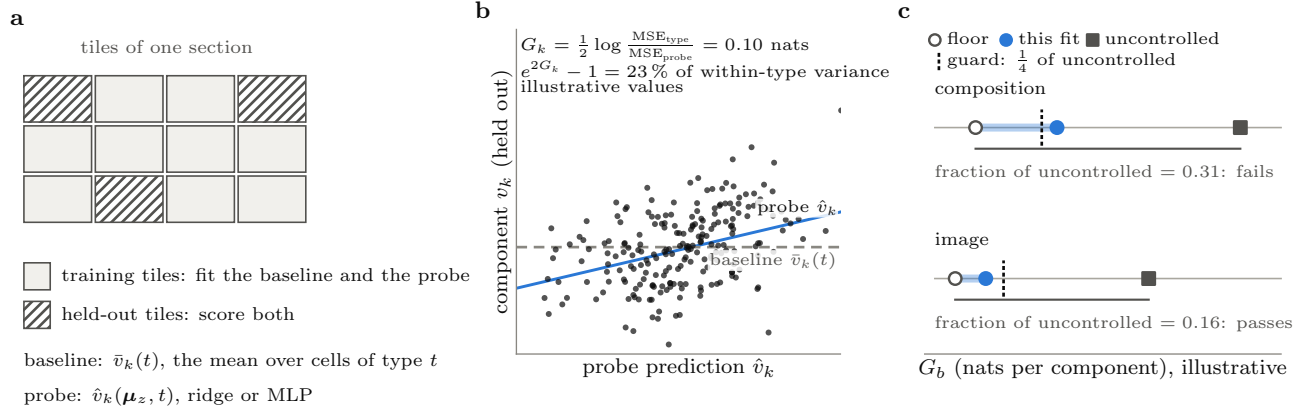


Figure 10: How the held-out probe grades the invariance of $\mathbf{z}$ (Equation (21)); schematic, with synthetic values chosen for legibility. **a**, The type-only baseline and the probe are fitted on the training tiles and scored on the held-out tiles (hatched). The baseline predicts each component of the invariance block by its mean over cells of the same type; the probe predicts it from (μ_z, t) with a ridge regression or a small perceptron. **b**, One component for held-out cells of one type. The baseline’s error is the spread around the type mean (dashed), the probe’s the spread around its prediction (solid, the line of perfect prediction). G_k is half the log ratio of the two mean squared errors, $e^{2G_k} - 1$ is the share of within-type variance the probe explains, and G_b averages G_k over the block. **c**, Reading G_b for each block. The floor is its value with μ_z permuted within type; the uncontrolled fit is the same model with $\alpha_a = 0$. What is reported is this fit’s excess over the floor (blue bar) and that excess as a fraction of the uncontrolled fit’s (grey bracket), for each probe family. A block passes when that fraction is at most a quarter (dotted guard): here the composition block fails and the image block passes.

gap, cannot be measured on tissue, where the exact posterior is unknown, so we measure it on simulated sections for which the exact posterior can be computed.

The simulation. Each simulated section follows DISCELL’s generative model with the response switched off, so that every quantity lies inside the model class the fit assumes and a gap measured here is a gap of inference, not of model mismatch.

- *Cells and types.* 3,000 cells at uniformly random positions in a $700 \mu\text{m}$ square, and four types laid out in spatial patches: each cell takes the type of the nearest of twelve anchor points, three per type, up to a random perturbation, so that neighbours are often of another type and leakage mixes types, as in tissue.
- *Intrinsic state and composition.* $\mathbf{z}_i \sim \mathcal{N}(\mathbf{m}_{t_i}, 0.5^2 \mathbf{I})$ in two dimensions, with type means $\mathbf{m}_t \sim \mathcal{N}(\mathbf{0}, 1.5^2 \mathbf{I})$, and a clean composition $\rho_i = \text{softmax}(\mathbf{z}_i \mathbf{A})$ over forty genes, with loadings $\mathbf{A}$ drawn from a standard normal.
- *Leakage.* The Delaunay graph of the positions, pruned at $40 \mu\text{m}$, with the kernel $\beta_{ij} \propto \exp(-d_{ij}/\tau)$, $\tau = 20 \mu\text{m}$, normalised over each cell’s neighbours: the model’s own kernel with every face set to one. The leak fraction is $\kappa = 0.2$, and $\bar{\rho}_i = \sum_j \beta_{ij} \rho_j$.
- *Counts.* A depth $\ell_i \sim \text{Poisson}(D)$, at least 20, for $D \in \{100, 300, 1,000\}$, and $\mathbf{x}_i \sim \text{Multinomial}(\ell_i, (1 - \kappa)\rho_i + \kappa\bar{\rho}_i)$.
- *Image descriptor.* An eight-dimensional vector made from the cell’s neighbour composition plus noise, so that the context has something to read.

The exact posterior. Given $\bar{\rho}_i$, κ , $\mathbf{A}$, $\mathbf{m}_{t_i}$ and ℓ_i , the posterior of $\mathbf{z}_i$ is a two-dimensional density. It has no closed form, since the multinomial and the softmax are not conjugate, but in two dimensions its mean and covariance are computed to numerical precision on a 129×129 grid centred on its mode.

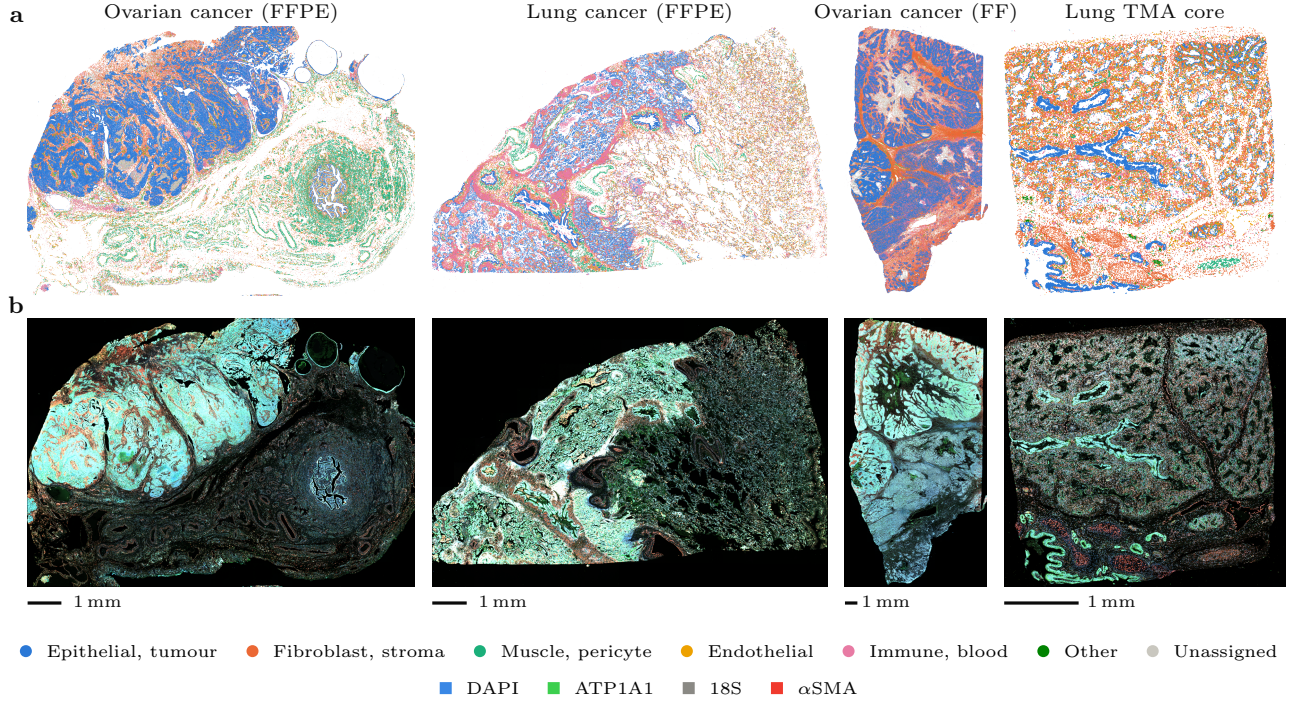


Figure 11: The four fitted sections. **a**, Every cell at its centroid, coloured by its lineage label, with the labels grouped into six families for display only; the model is fitted with the finer lineage classes of Table 9. **b**, The morphology image of the same section, the four channels the image descriptor reads, coloured as in Figure 1b. The TMA panel is the fitted core only. Each section is shown at its own scale; the bars are 1 mm.

The fit and the comparison. DISCELL is fitted to the counts with a two-dimensional $\mathbf{z}$ and its response channel present, as on tissue, for 300 epochs, at the true leak fraction and at leak fractions off by up to 0.1 either way. Because $\mathbf{z}$ is identified only up to a smooth invertible map, the encoder’s posterior means are first mapped onto the simulation’s coordinates by a small network with five-fold cross-fitting; since the means are two-dimensional, the map cannot add information the encoder did not keep. The gap of a cell is the distance between its mapped mean and the exact mean, divided by the size of the exact posterior’s standard deviation, and it is averaged over cells and over three seeds. Four references are fitted to the exact posterior mean directly, also with cross-fitting: a multilayer perceptron given the encoder’s inputs, the same perceptron given $\bar{\rho}_i$ as well, which the encoder never sees, a linear regression on the encoder’s inputs, and the type mean, which knows no counts.

The result. At the true leak fraction the encoder’s gap is below that of the perceptron trained directly on the same inputs, at every depth, and close to that of the perceptron that also sees $\bar{\rho}_i$ (Table 14): not seeing the influx costs little. The gap grows with depth only in units of the posterior width, because the posterior narrows as the depth increases; in units of $\mathbf{z}$ itself it is flat. A wrong leak fraction enlarges it by at most a tenth over the mismatches tried, and it is smallest at the true value. The references are regression models of finite capacity, not information bounds, so these statements are comparative.

A leak-subtracted encoder input. The intrinsic encoder could be given counts from which the expected foreign influx has been subtracted, rather than the raw counts. On planted worlds with a known leak, this helped on one of three worlds. In a follow-up with twelve seeds it reversed on three of the six seeds where the comparison was not saturated, and it cost up to twenty-two points of recall of the planted signal. The encoder therefore reads raw counts, and the subtraction is kept as an option, off by default.

A dead response channel. A fit can end with a response that carries no information about the niche: the divergence of $\mathbf{w}$ to its prior collapses and the context is ignored. Two checks flag it. During training, a summed divergence of $\mathbf{w}$ below 10^{-5} over the first twenty epochs marks the fit. After training, the information between $\mathbf{w}$

and the niche, on held-out cells and against a within-type permutation floor, must exceed its floor. Among the archived fits of the development period, twelve had a dead channel, nine of them on the deepest section. The KL warm-up of the final configuration removed it in every one of eight refits of that section, and the checks flag no final fit.

Table 5: Networks and fixed quantities. G genes, K types, d_Φ image-embedding dimension.

Component	Input	Layers / value
Count encoding	$\mathbf{x}_i$	$[\log(1 + \mathbf{x}_i \tilde{\ell} / \ell_i), \log \ell_i]$, $\tilde{\ell}$ the median total
$q(\mathbf{z}_i \mathbf{x}_i, t_i)$	count encoding $\oplus$ onehot(t_i)	$256 \rightarrow 256 \rightarrow 2d_z$, $d_z = 20$
Type query embed(t_i)	t_i	learned table, width K
Attention sources	neighbour j	onehot(t_j), width K ; an earlier version appended the neighbour's posterior mean under a stop-gradient and was dropped after a three-seed ablation
Attention layer	sources, query	one bipartite GATv2 layer, 4 heads $\times$ 32, averaged, LeakyReLU slope 0.2, no self-loops, no edge features
Context $\mathbf{c}_i$		attention output (32) $\oplus \Phi_i$ ($d_\Phi = 384$) $\oplus$ isolation flag (1)
Prior mean $m_\psi(\mathbf{c}_i, t_i)$	$\mathbf{c}_i \oplus \text{onehot}(t_i)$	$64 \rightarrow d_w$, $d_w = 6$
$q(\mathbf{w}_i \cdot)$	$\mathbf{c}_i \oplus \text{onehot}(t_i) \oplus \mathbf{z}_i \oplus$ count encoding	$256 \rightarrow 2d_w$
Decoder $a(\mathbf{z}_i)$	$\mathbf{z}_i$	$256 \rightarrow G$
Loadings $\mathbf{B}$	$\mathbf{w}_i$	linear $d_w \rightarrow G$, no bias
Adversary heads	$\mu_{z,i} \oplus \text{onehot}(t_i)$	two heads $64 \rightarrow 64 \rightarrow K$ (composition) and $\rightarrow E_\Phi = K$ (image niche)
Image niche e_i^Φ	12 principal components of Φ	soft k -means memberships softmax($-d^2/T$), T the median nearest squared distance; the projection is fitted on a 50,000-cell random subsample and applied to all cells, the k -means (four restarts) on all cells; fitted once and frozen
Invariance block $\mathbf{v}_i$		$[\mathbf{y}_i^{(-K)}, 12 \text{ principal components of } \Phi_i]$
Held-out probes	$\mu_{z,i} \oplus \text{onehot}(t_i)$	ridge regression ($\lambda = 10^{-3}$) and perceptron $64 \rightarrow 64$ (300 Adam steps at 10^{-3} , batches of 2,048), each scored per block against its own within-type permutation floor; both fitted on at most 30,000 training-tile cells
Graph		Delaunay adjacency; Voronoi faces measured within $30 \mu\text{m}$ discs; prune at $40 \mu\text{m}$
Kernel β		$f_{ij} \exp(-d_{ij}/\tau)$, $\tau = 20 \mu\text{m}$, row-normalised over in-edges
Prior scale σ_w		1 (fixed)
Leak fraction κ		swept over $\{0, 0.05, 0.1, 0.2, 0.3, 0.4\}$; the operating point 0.1 is a grid point chosen on the primary section (Section 3.1)
ω		1
α_z		$\frac{1}{2}/\bar{\ell}$, half the bound-equivalent value, with $\bar{\ell}$ the section's count scale (Section 2.4)
α_w		0.1, far above $1/\bar{\ell}$ (Section 2.4); ramped linearly from 0 over the first 30 epochs
α_a (adversary)		0.3; six head updates per model update at learning rate 2×10^{-3} , set so that the independent probe, not the heads' own loss, certifies head strength; composition half weighted $\lambda_y = 3$
α_a (closed form)		calibrated by the same probe protocol
Moving-average shrinkage	rate,	0.05, 0.05; support floor $10 \times$ joint dimension
Tiles		recursive median split into rectangles of $\leq 4,096$ cells; 15% of tiles held out
Optimiser		Adam (Kingma and Ba, 2015), learning rate 10^{-3} , cosine annealing (Loshchilov and Hutter, 2017) over the epoch budget, gradient-norm clip 10
Evaluation		every 5 epochs on posterior means; checkpoint accepted if held-out reconstruction improves and $\mathbf{z}$ -type agreement $\geq 0.9 \times$ its running maximum; patience in epochs set with the budget; α_w ramped linearly from 0 over the first 30 epochs, during which no checkpoint is accepted and patience does not count
Hardware		one NVIDIA GeForce RTX 4090 (24 GB) per fit

Table 6: Implementation choices and their reasons.

Where	Decision	Reason
Type input to $q(\mathbf{w})$ and m_ψ	one-hot everywhere the type is an input; the learned type embedding exists only as the attention query, sized to match the source features	the query stands in for the $\mathbf{z}_i$ that the cell deliberately does not contribute to its own context
Counts input to $q(\mathbf{w})$	the same encoding as $q(\mathbf{z})$	one encoding of the counts for both posteriors
Neighbour responses $\mathbf{w}_j$	a posterior draw from the same batched pass, under a stop-gradient	the neighbours' clean compositions must include their local programme (Section 2.6)
Invariance block	one composition column dropped	the simplex makes the full block singular; the information is unchanged (Appendix A.4)
Penalty matrices	correlation rather than covariance form, plus $10^{-5}\mathbf{I}$ on the correlation matrix before the log-determinant	identical value by scale invariance; bounded gradient; on a unit-diagonal matrix the additive term is a uniform relative jitter, not a variance-inflating ridge
Penalty gradient	straight-through: value from the moving averages, gradient from the batch	otherwise the effective strength of α_a scales with the averaging rate
Penalty, adversary and probe input	posterior means $\boldsymbol{\mu}_z$, not samples	encoder noise would dilute the measured dependence
Objective granularity	per-cell mean over the batch rather than a sum	invariant to tile size
Evaluation draws	posterior means	the stopping signal should not ride reparameterisation noise
Image embedding dimension	the full native embedding into $\mathbf{c}_i$ rather than a 32–64-dimensional projection	the deciding test is held-out reconstruction with and without the projection; the invariance block uses 12 principal components regardless
Probe target for the image	Gaussian log-likelihood gain on the continuous components, per block, rather than categorical cross-entropy against image-niche clusters	the closed-form penalty takes continuous components directly; a small nonlinear probe is fitted alongside and is the probe the decision rule reads (Section 2.5)
Graph construction	the Voronoi tessellation is clipped to $30\mu\text{m}$ discs before faces are measured, then edges are pruned at $40\mu\text{m}$	the clip removes faces only for pairs more than roughly $60\mu\text{m}$ apart, which the prune removes anyway, and bounds the length of faces at tissue borders and across gaps; the effect on β for kept edges has not been quantified [TODO: quantify]
Invariance mechanism	the adversary is the operating configuration; the closed-form penalty is the cheaper first attempt	the escalation rule of Section 2.5 decides between them on each section

Table 7: Implementation choices, continued: the configuration used in the experiments, and what exists but is switched off.

Where	Decision	Reason
Attention sources	neighbour types only; an earlier version also passed the neighbour’s posterior mean under a stop-gradient	the case is empirical, not mechanistic: a seed-matched ablation improved held-out reconstruction, the $\mathbf{z}$ -type agreement, the mirror check and the cycle readout of $\mathbf{w}$ while leaving $\mathbf{w}$ ’s spatial readouts unchanged. The mechanism originally proposed for it, the prior reconstructing a cell from its neighbours’ codes, was tested with two purpose-built detectors and not confirmed; the sources are types because the ablation is better, not because that mechanism was demonstrated
Training budget	two budgets: a short one for sweep points, a long one for reference fits (Section 2.7)	sweep points must be comparable to one another; reference fits must be individually good, and the better optimisation basin is not reached at the short budget
Choosing among seeds	several seeds per reference configuration, one selected on the full diagnostic battery, the rest reported as the envelope	selection on held-out reconstruction alone admits a fit that buys likelihood at the cost of the intrinsic-state readouts
Halo overhead	ring one is fully encoded; ring two contributes a type lookup only	the attention sources are types, so ring two needs no encoder pass and none is run; the node overhead falls with tile size, and tiles are sized above the knee
Resident count storage	the count matrix is held on the accelerator as 16-bit integers and cast per tile	counts are small non-negative integers, so the narrow type is exact and is asserted to be; the wide type does not fit on one consumer card for the largest section, and tile size does not help, since every tile is resident
Built but not default	leak-subtracted encoder input $\tilde{\mathbf{x}}_i = \max(\mathbf{x}_i - \kappa \ell_i \bar{\rho}_i, 0)$, normalised by its own total with the log-depth scalar left raw, as the counts-level answer to the amortisation gap; an attention sink, a fixed null logit joining every destination’s softmax so that the aggregate saturates in the number of neighbours instead of ignoring it; coupled weight decay in the optimiser	each is available behind a flag and off in every fit reported here. The encoder correction is retained as an ablation rather than as the model, since it changes the evidence $\mathbf{z}$ is formed from; the sink cannot be exercised on a Delaunay graph, whose degree is nearly constant, and is kept for graphs whose degree varies; weight decay is recorded because it moves the gauge of Section 2.8 rather than removing it
Not built	Dirichlet-multinomial likelihood; per-cell counterfactuals with abduction; bounded conditional $\sigma_w(\mathbf{c}, t)$; estimation of κ from a nuclear/extranuclear split; edge features in the attention layer	each has a stated trigger that has not fired, or is future work. The <i>type-level</i> counterfactual, in which the context prior is moved between two neighbourhoods with the intrinsic state held fixed, is built and exercised; only the per-cell form, which needs a non-degenerate response posterior, is not

Table 8: The held-out probe on the final fits, per section. For each block, the share of its within-type variance that the probe explains from $\mathbf{z}$ beyond the permutation floor ($e^2_{\text{excess}} - 1$, in %), for the model trained without the adversary ($\alpha_a = 0$, two seeds) and with it (the final fits, three seeds), the largest share with the adversary that would pass (*needed to pass*: a quarter of the excess without the adversary, as a share), and the quantity the guard reads, the *fraction left*: the excess with the adversary as a fraction of the mean excess without it, 0 when the adversary removes everything the probe can find and 1 when it removes nothing. Ranges are over seeds. A seed passes when all four fractions (two blocks, two probes) are at most a quarter. The sections end at a similar absolute residual; the fraction left is larger where less leaked to begin with. Section B of the lung TMA is graded with the models fitted on section A. **[PENDING RUNS: regenerate once Unassigned is excluded as a target in the re-read of the final fits]**

	Ovarian FFPE	Lung FFPE	Ovarian FF	Lung TMA A	Lung TMA B
<i>Composition, MLP probe</i>					
without the adversary (%)	19.5–20.8	11.8–12.4	36.3–37.4	5.5–7.0	6.8–7.0
with the adversary (%)	4.4–5.7	3.9–4.7	4.4–4.9	3.1–4.4	3.9–4.6
needed to pass (%)	4.7	2.9	8.2	1.5	1.7
fraction left	0.23–0.30	0.33–0.40	0.14–0.15	0.50–0.71	0.58–0.67
<i>Image, MLP probe</i>					
without the adversary (%)	14.8–17.1	9.5–10.9	34.3–37.0	5.8–8.3	5.3–7.2
with the adversary (%)	2.5–4.1	1.4–2.0	4.5–4.9	2.2–2.8	2.0–3.0
needed to pass (%)	3.8	2.5	7.9	1.7	1.5
fraction left	0.17–0.27	0.14–0.20	0.14–0.16	0.32–0.40	0.32–0.49
<i>Ridge probe, fraction left</i>					
composition	0.16–0.23	0.15–0.19	0.07–0.08	0.19–0.22	0.32–0.45
image	0.08–0.16	0.05–0.16	0.10–0.13	0.19–0.21	0.17–0.39
Seeds passing	1/3	0/3	3/3	0/3	0/3

Table 9: The sections. Median counts are per cell over the whole section. The TMA core and its serial section are one core of the microarray on two slides; no model is fitted to the serial section.

Section	Role	Cells	Genes	Lineage classes	Unassigned (%)	Median counts
Ovarian cancer (FFPE)	primary	407,120	5,101	12	10.3	178
Lung cancer (FFPE)	fitted	278,324	5,001	20	0.1	242
Ovarian cancer (FF)	fitted	1,157,637	5,001	10	4.3	1,401
Lung TMA core	fitted	69,422	5,001	30	4.2	279
Lung TMA core, serial section	held out	70,757	5,001	30	4.8	253

Table 10: Lineage labels on the ovarian FFPE section: the curated classes merged into each lineage.

Lineage	Curated classes
Ciliated epithelial cells	Ciliated Epithelial Cells
Endothelial cells	Stromal Associated Endothelial Cells, Tumor Associated Endothelial Cells
Fallopian tube epithelium	Fallopian Tube Epithelium
Fibroblasts	Stromal Associated Fibroblasts, Tumor Associated Fibroblasts
Macrophages	Macrophages
Mesothelial-like cyst lining	Malignant Cells Lining Cyst
Pericytes	Pericytes
Smooth muscle cells	Smooth Muscle Cells
T/NK cells	T and NK Cells
Tumour	Inflammatory Tumor Cells, Proliferative Tumor Cells, Tumor Cells, VEGFA+ Tumor Cells
Unassigned	SOX2-OT+ Tumor Cells, Unassigned
Urothelial-like cells	Urothelial-like Cells

Table 11: Lineage labels on the two clustered sections: the clusters assigned to each lineage.

Lung cancer (FFPE)		Ovarian cancer (FF)	
Lineage	Clusters	Lineage	Clusters
AT1	18, 23	Endothelial cells	16, 38
AT2	20	Fibroblasts	7, 13
Adventitial fibroblasts	4, 17	Macrophages	10, 14, 19, 20, 25, 26, 28, 30, 34
Alveolar fibroblasts	10	Ovarian stroma	3, 6, 23
Alveolar macrophages	16	Pericytes	32
B cells	12	Plasma cells	17, 35
EC general capillary	3, 22, 25	Smooth muscle cells	33
EC venous	13	T/NK cells	18, 22, 24, 27, 29, 31, 36, 37
Goblet	19	Tumour	1, 2, 4, 5, 8, 11, 12, 15, 21
Interstitial macrophages	2	Unassigned	9; unclustered cells
Mast cells	27		
Multiciliated	26, 28		
Neuroendocrine	31		
Neutrophils	6		
Pericytes	24		
Plasma cells	15		
Smooth muscle	5		
T/NK cells	1, 7, 21, 30, 32		
Tumour	8, 9, 11, 14, 29		
Unassigned	unclustered cells		

Table 12: Comparison methods as run. Versions are the package versions used.

Method	Version	Graph	Label	Budget and coverage
resolVI	scvi-tools 1.5.1	own, 10 nearest neighbours	lineage	50 epochs; every section package default; the TMA core and its serial section; failed on the fresh-frozen section (the whole section does not fit in GPU memory); not attempted on the two FFPE sections
SIMVI	simvi 0.1.2 (scvi-tools 0.16.1)	ours	not used	
MintFlow	mintflow 0.3.0	own Delaunay graph	lineage	
Cellina	cellina 1.1.1 (scvi-tools 1.2.0)	ours	lineage	width 600, 50 epochs; the TMA core, transferred to its serial section; failed on the fresh-frozen section (out of host memory before training); not attempted on the two FFPE sections package default; being run

Table 13: Model size and training cost on one GPU. *Peak memory*: largest allocation during training, with the section resident on the GPU. *Run time*: one whole fit, for DISCELL including its evaluations every five epochs and the early stop, as the range over the three seeds of the final configuration. Where a method has no time, the reason is given: *not attempted* means it was never run on that section, *failed* that it was run and did not complete.

	Ovarian (FFPE)	Lung (FFPE)	Ovarian (FF)	Lung TMA core
<i>DISCELL, model and training details</i>				
parameters (M)	4.19	4.12	4.11	4.13
peak memory (GiB)	6.3	4.3	15.1	1.4
epochs to the stop	105–115	80–100	175–230	85–90
s per epoch, training only	1.98	1.35	5.78	0.45
<i>Run time of one fit (min)</i>				
DISCELL	8.7–9.3	3.2–7.4	31.2–43.4	1.3
resolVI	32.8	22.0	105.0	5.5
SIMVI	not attempted ^a	not attempted ^a	failed ^b	107.2
MintFlow, 50 epochs	not attempted ^c	not attempted ^c	failed ^d	680.1

^a Run only on the TMA core, where it fits whole; not attempted on the two FFPE sections. ^b SIMVI passes the whole graph through the network at every step, and the whole fresh-frozen section does not fit in GPU memory. We report only methods run on whole sections, so an earlier run on a window of the section is not used. ^c Run only on the TMA core, where one fit took about eleven hours for 70,000 cells; not attempted on the larger FFPE sections. ^d Ran out of host memory while building its inputs, before training started.

Table 14: Amortisation gap on the simulated sections at the true leak fraction: the distance between an estimate of the posterior mean of $\mathbf{z}$ and the exact one, in units of the exact posterior standard deviation, averaged over cells, mean of three seeds, by sequencing depth D . The DISCELL encoder is trained without the exact posterior; the other rows are fitted to it directly.

	Depth D		
	100	300	1,000
Estimate of the posterior mean			
DISCELL encoder	0.55	0.83	1.48
Perceptron, the encoder’s inputs	0.61	0.95	1.61
Perceptron, the encoder’s inputs and $\bar{\rho}_i$	0.53	0.80	1.35
Linear regression, the encoder’s inputs	1.40	2.50	4.53
Type mean	2.4	4.1	7.4